

Modeling of arterial wave propagation: theory & applications

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Course content

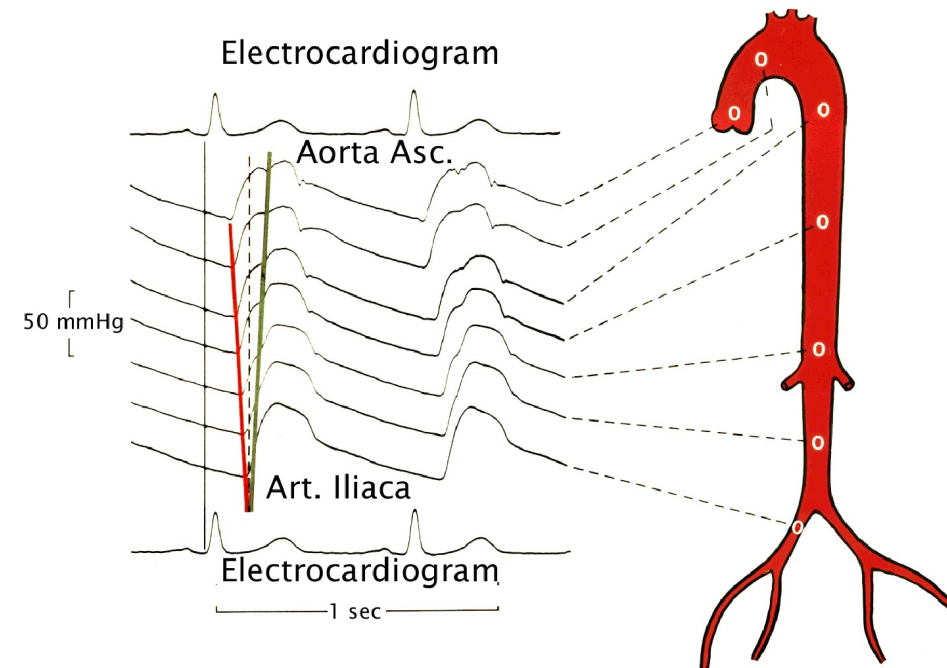
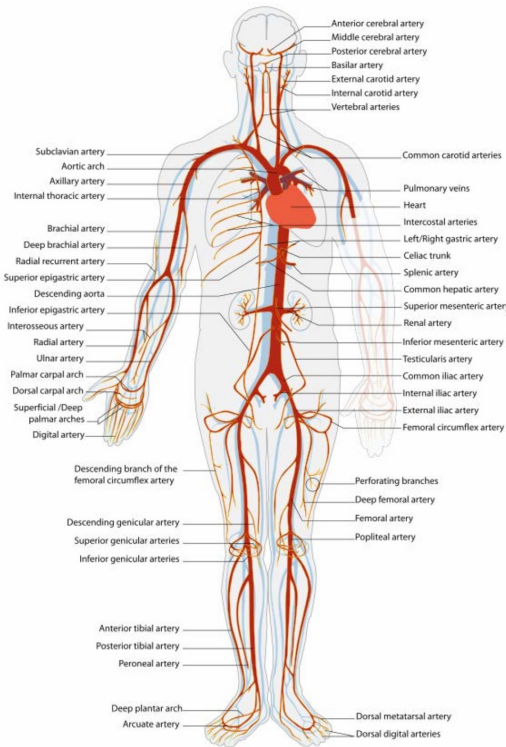
Theoretical elements

- Notions of pressure and flow wave propagation and clinical importance
- 1-D flow models
- Pulse wave velocity and characteristic impedance
- Global models (2-, 3- and 4-element Windkessel models)

Applications

- Inverse methods for estimation of central hemodynamical parameters
 - Cardiac output
 - Cardiac contractility
- Virtual patients for ML

Wave propagation in the arterial system

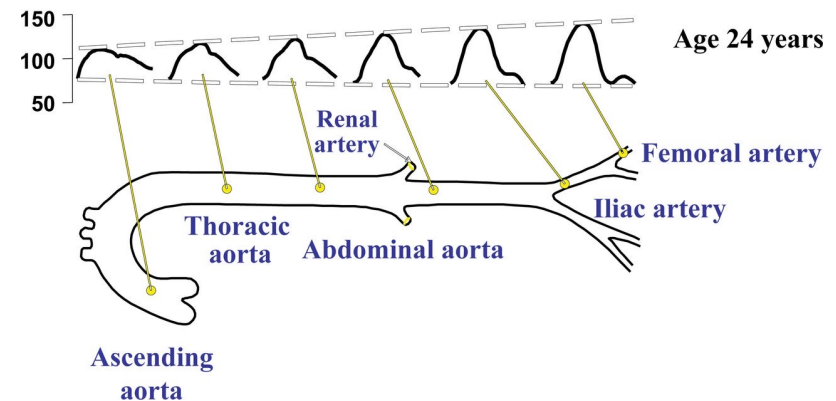


Arterial system:

- Complex network of viscoelastic tubes
- Complicated geometry
- Nonlinear elastic properties
- Non-newtonian fluids, large variation of Reynolds numbers

Aortic BP curve:

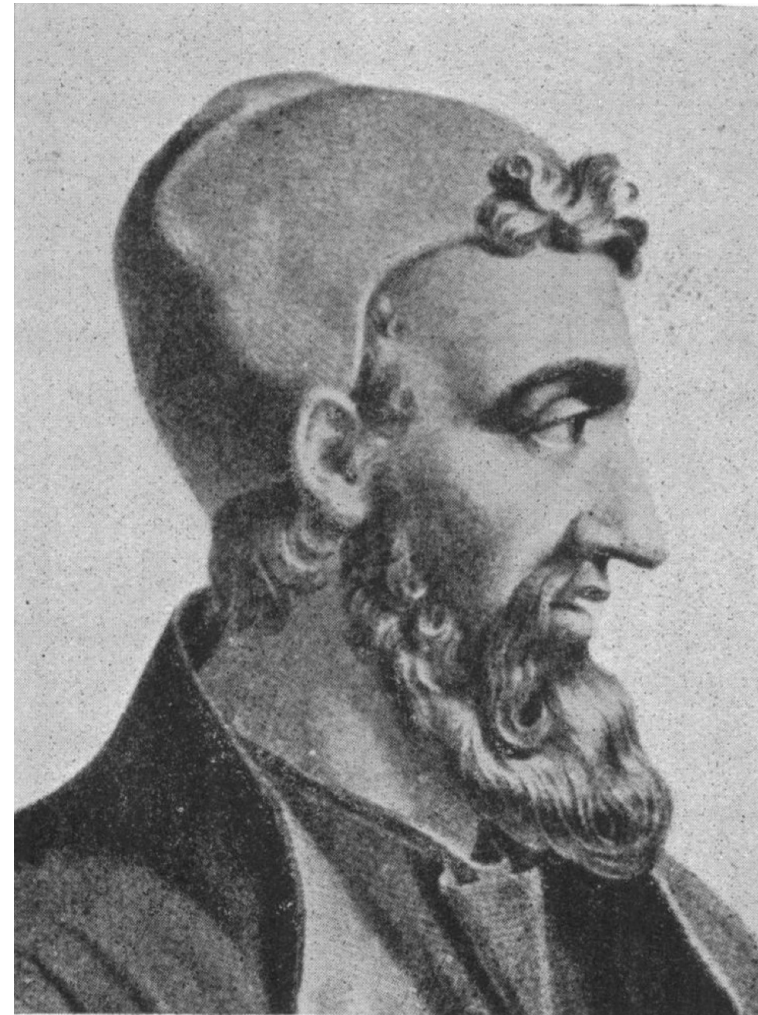
Propagation and amplification



Feeling the pulse...

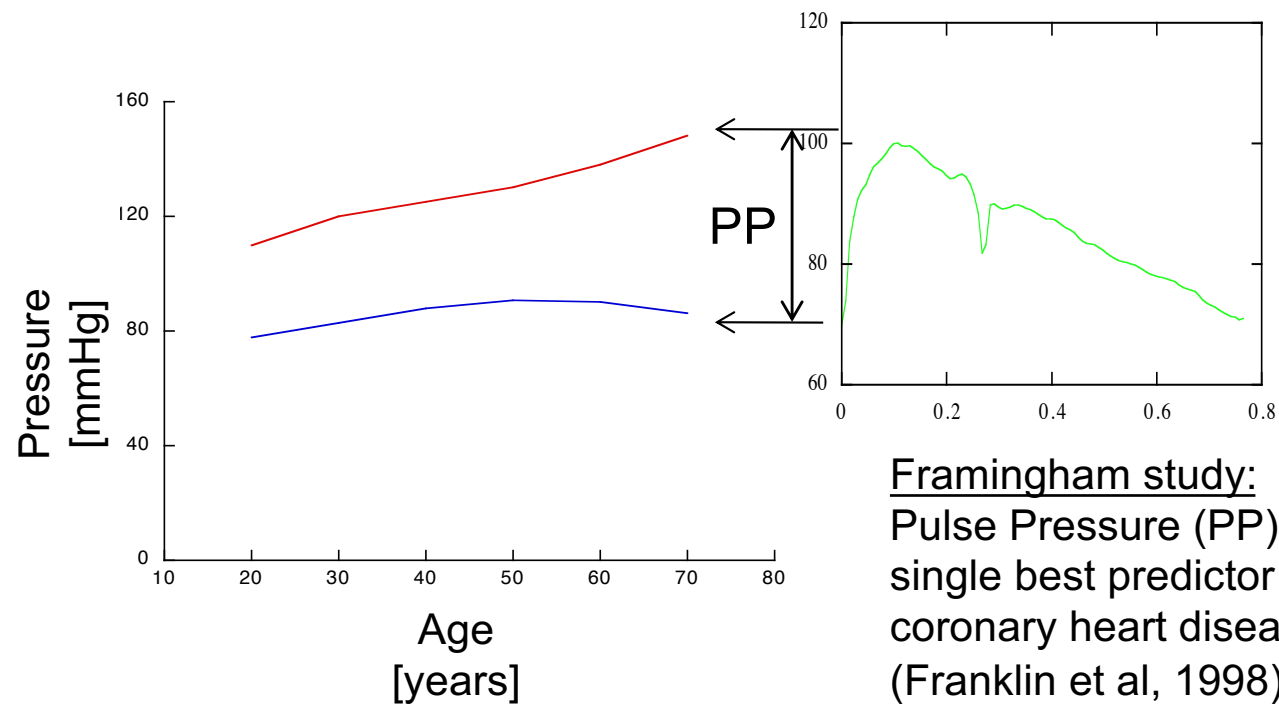


A consultation by Susruta, the Hindu physician, who is portrayed feeling the pulse

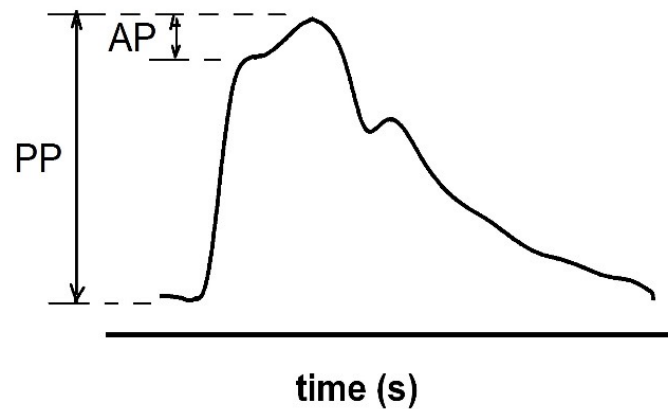
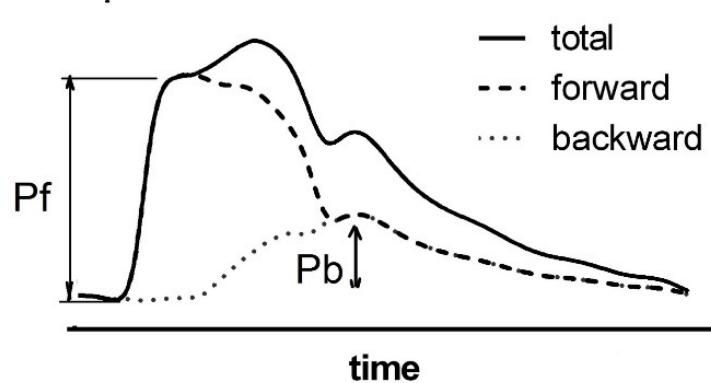


Claudius Galenus (A.D.131-200). The greatest sphygmologist of antiquity and author of 18 books on the arterial pulse

Hemodynamics and Cardiovascular Ageing



Augmentation index

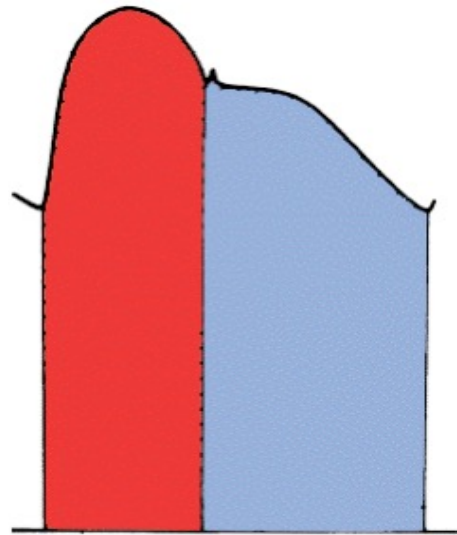


Augmentation index
 $AI = AP/PP$

Aortic pressures in young and old subjects

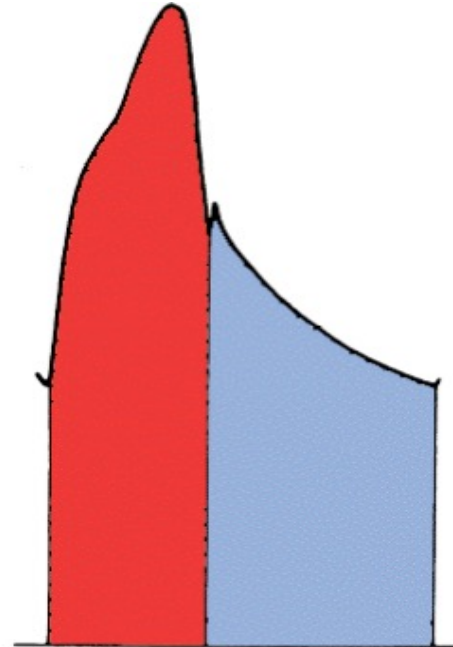
Young

Type C

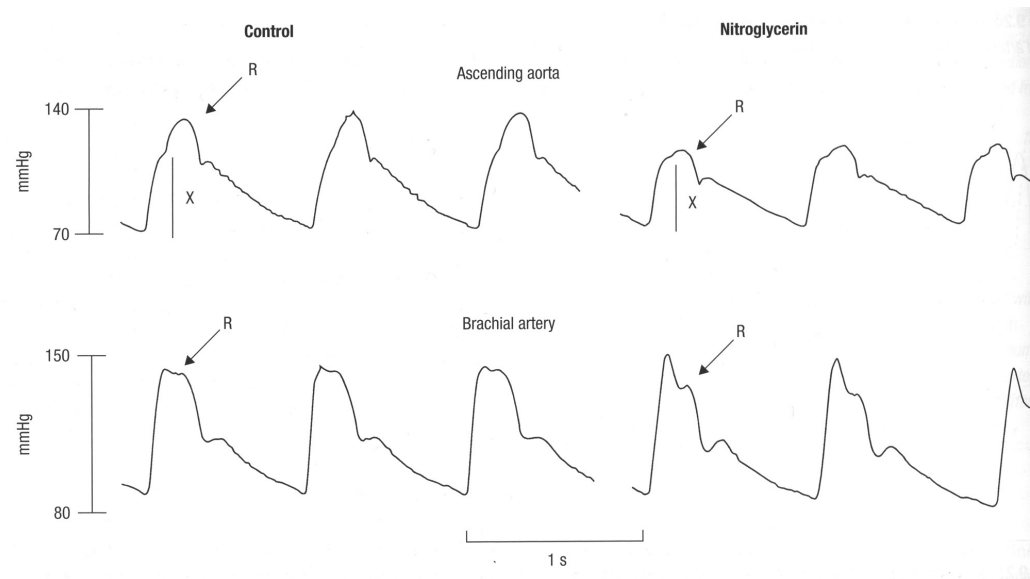


Old

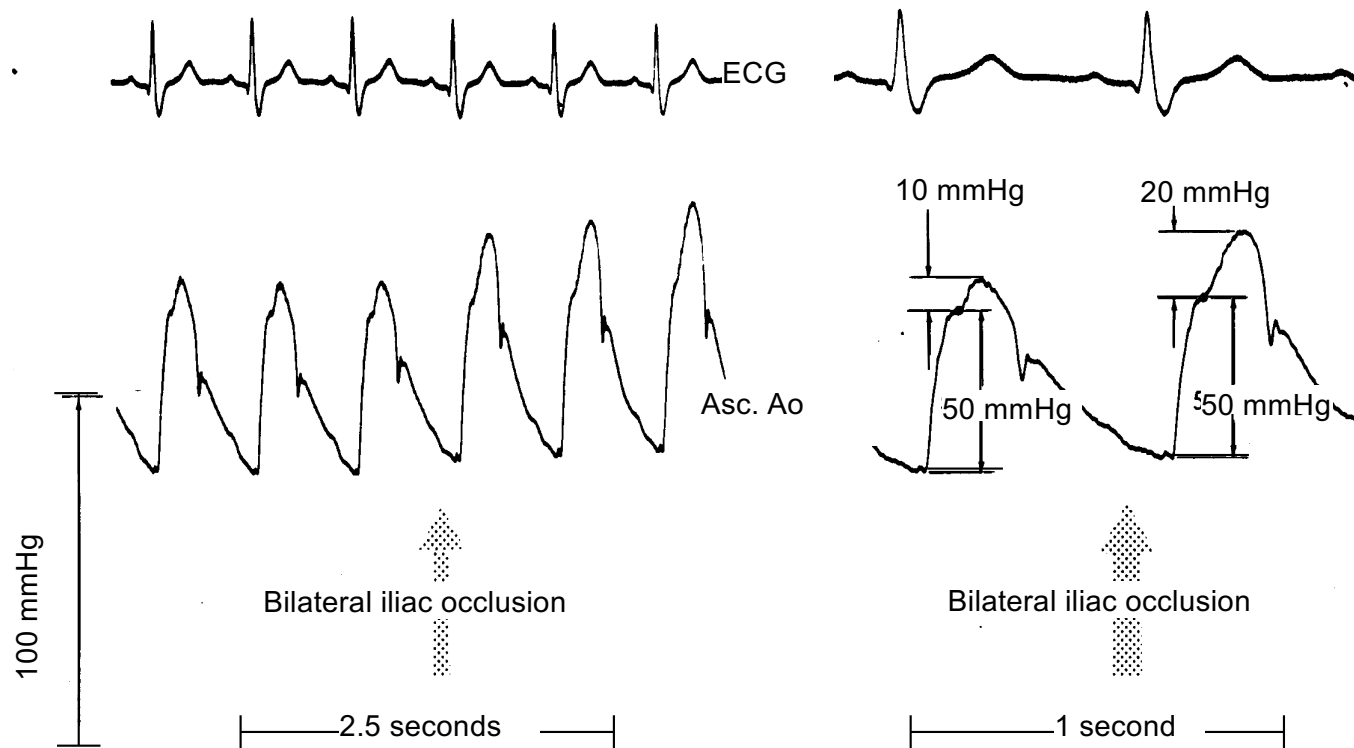
Type A



Clinical implications: monitoring pressure

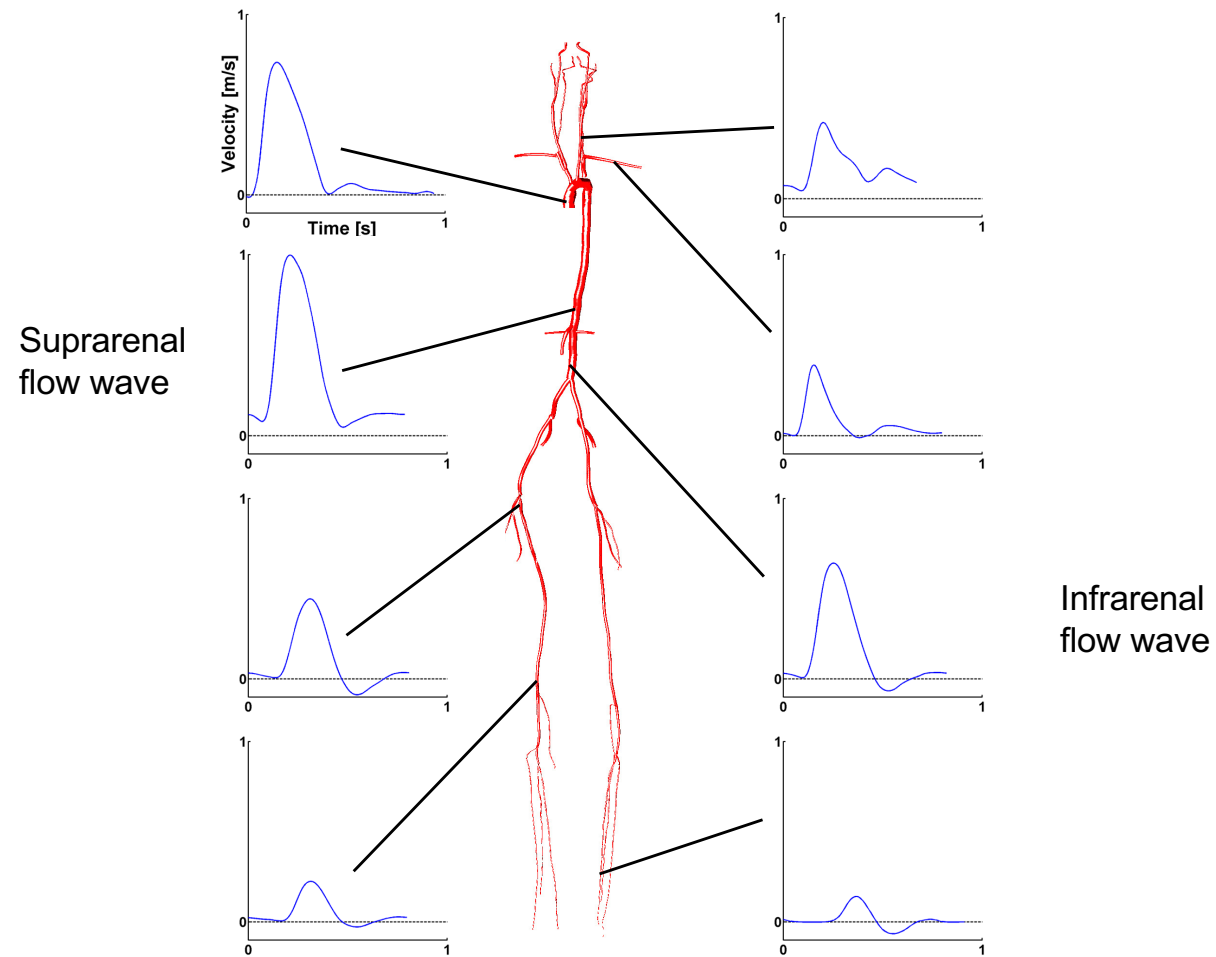


Wave reflections: augmentation by peripheral occlusion

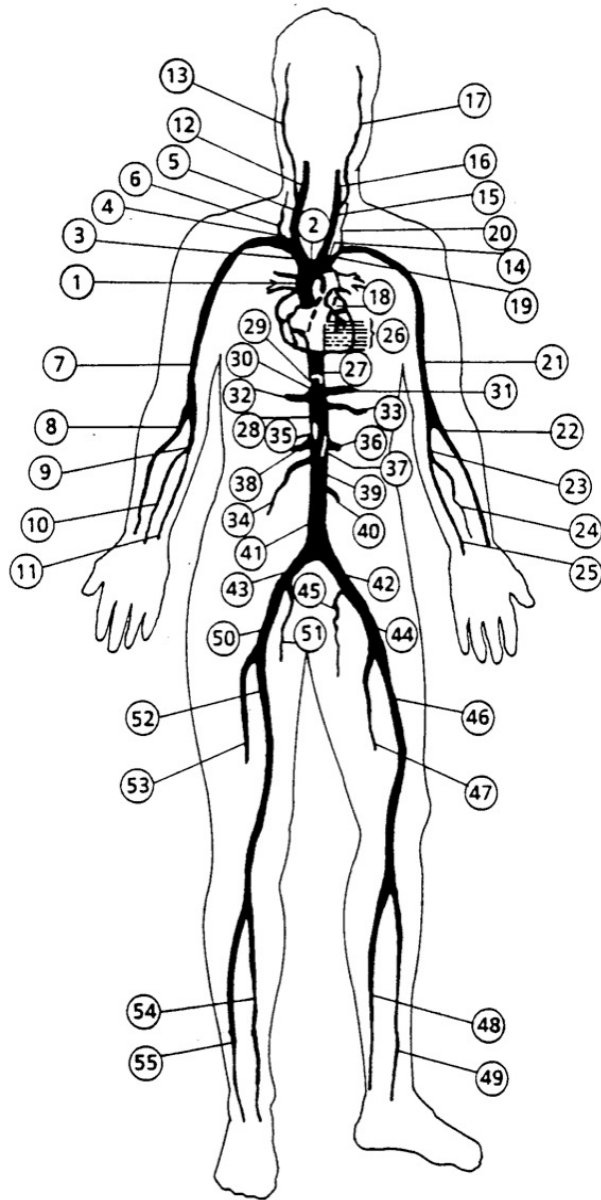


DISTINCT REFLECTION IS INCREASED by mechanical compression of both iliac arteries.

Flow wave propagation of importance as well!



1-D model of the arterial system



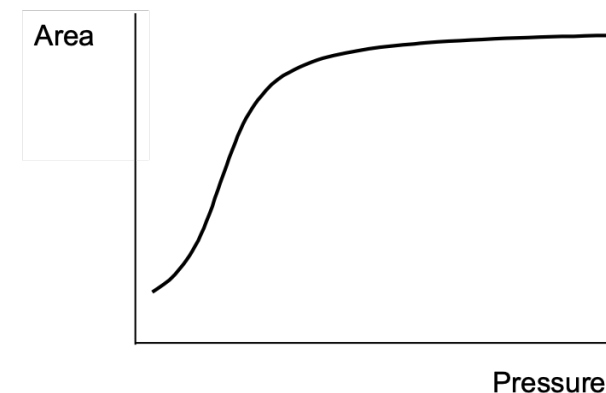
Continuity

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0$$

Momentum

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x} \left(\lambda \frac{Q^2}{A} \right) = - \frac{A}{\rho} \frac{\partial P}{\partial x} + \frac{\pi D \tau_w}{\rho}$$

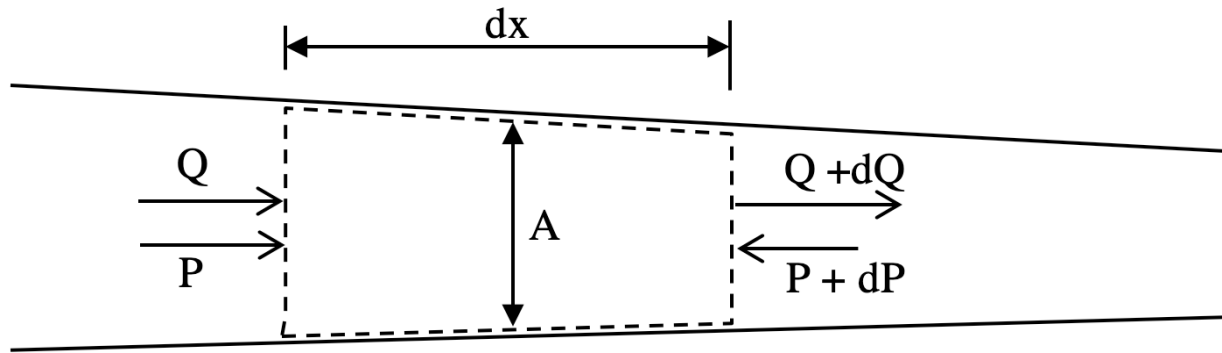
Constitutive law



- Three equations (Continuity, momentum and constitutive law $A=f(p)$)
- Three unknowns:
 1. Area (A),
 2. Pressure (P) and
 3. Flow (Q)

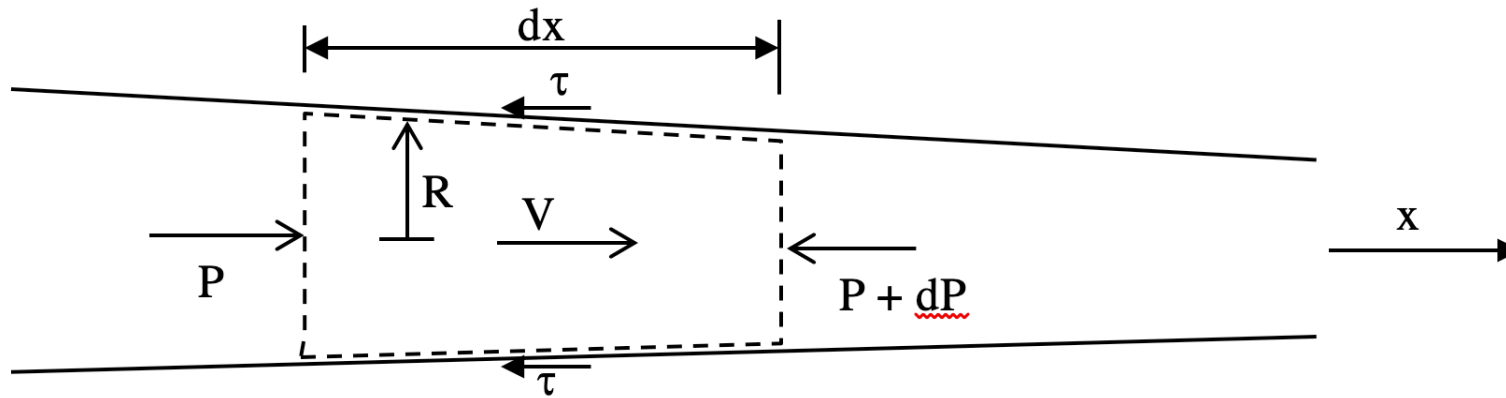
1-D model of the arterial system

Conservation of mass



$$Q_{in} - Q_{out} = \frac{dV}{dt} \Rightarrow Q - (Q + dQ) = \frac{d(dx \cdot A)}{dt} = dx \frac{\partial A}{\partial t} \Rightarrow \underline{\underline{\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0}} \quad (1)$$

Conservation of momentum



Assumption: no radial velocity component ($v \ll u$)

Newton's 2nd law:

$$\sum F_x = m \cdot a_x$$

$$\Rightarrow [P - (P + dP)] \cdot A - \tau \cdot (2\pi R \cdot dx) = (\rho \cdot A \cdot dx) \frac{\partial V}{\partial t}$$

Note: $Q = A V$

$$dQ = d(A \cdot V) = dA \cdot V + A \cdot dV \approx A \cdot dV$$

$$\Rightarrow -\frac{\partial P}{\partial x} \cdot A - \tau \cdot 2\pi R = \rho \cdot \frac{\partial Q}{\partial t}$$

$$\Rightarrow \underline{\underline{\frac{\partial Q}{\partial t} = -\frac{A}{\rho} \frac{\partial P}{\partial x} - \tau \cdot \frac{2\pi R}{\rho}}} \quad (2a)$$

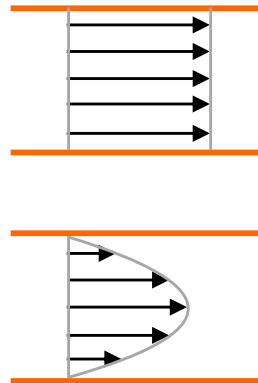
We have neglected convective acceleration (tapering) effects. If included, Eq. (2a) becomes:

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x} \left(\lambda \frac{Q^2}{A} \right) = - \frac{A}{\rho} \frac{\partial P}{\partial x} - \tau \cdot \frac{2\pi R}{\rho} \quad (2b)$$

Problem:

$$\left. \begin{aligned} \lambda &= \frac{1}{A} \int \frac{u^2}{V^2} dA \\ \tau &= \mu \left. \frac{\partial u}{\partial r} \right|_{r=R} \end{aligned} \right\} \text{ both depend on the velocity profile shape factor.}$$

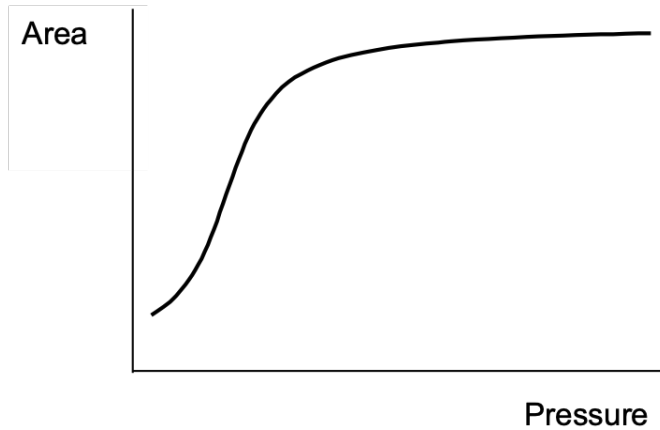
$$\lambda \left\{ \begin{aligned} &= 1 \text{ for a flat velocity profile} \\ &= 4/3 \text{ for a parabolic velocity profile} \end{aligned} \right.$$



The wall shear stress is usually approximated by Poisseuille's law: $\tau = \frac{4\mu Q}{\pi R^3}$

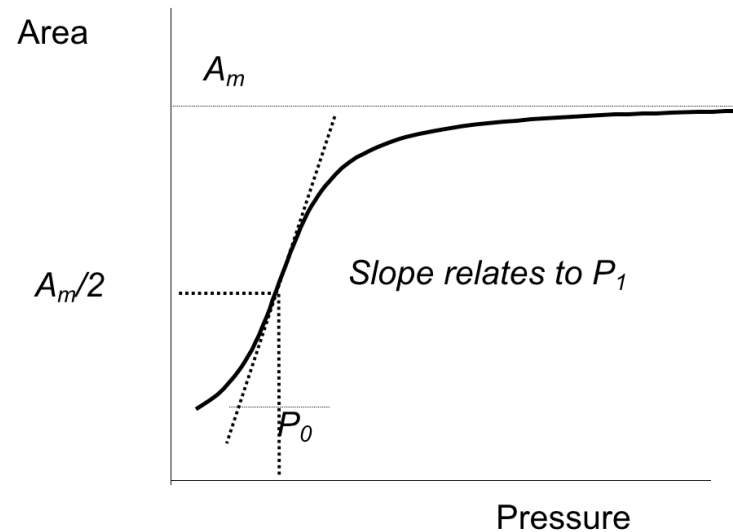
Constitutive equation

A constitutive equation gives the relationship between pressure and cross-sectional area: $A = f(p)$.



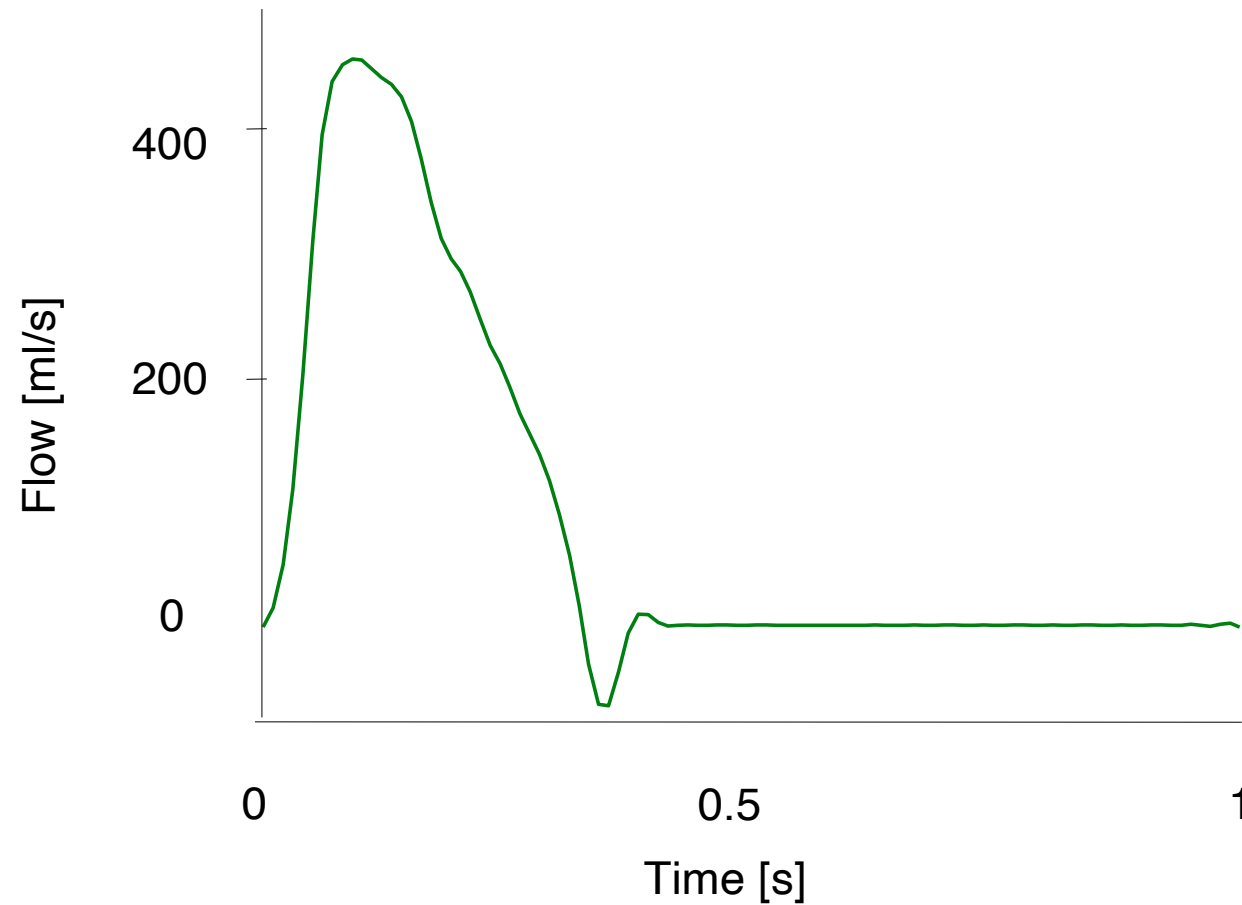
In analytical form, we may use the Langewouters formula:

$$A = A_m \left[\frac{1}{2} + \tan^{-1} \left(\frac{P - P_0}{P_1} \right) \right]$$

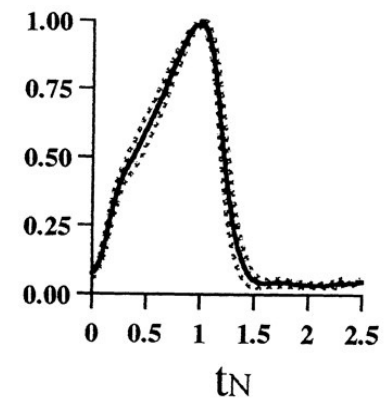
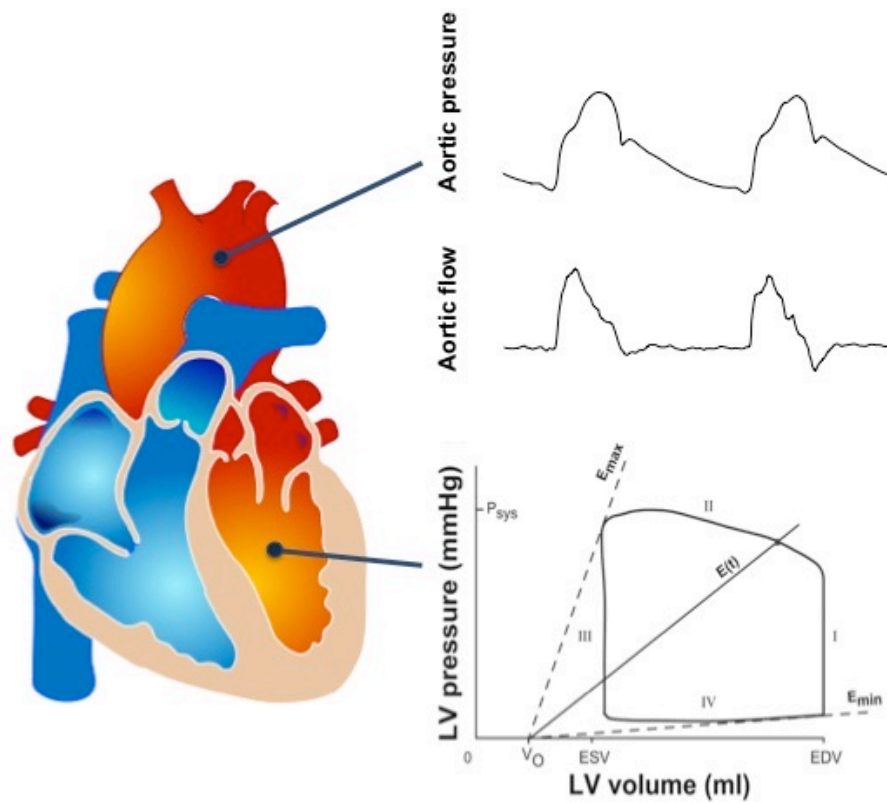


Upstream boundary condition

- Specify either the (aortic) pressure or flow waveform.
- Most often the aortic flow waveform (figurer below) is used as the upstream boundary condition

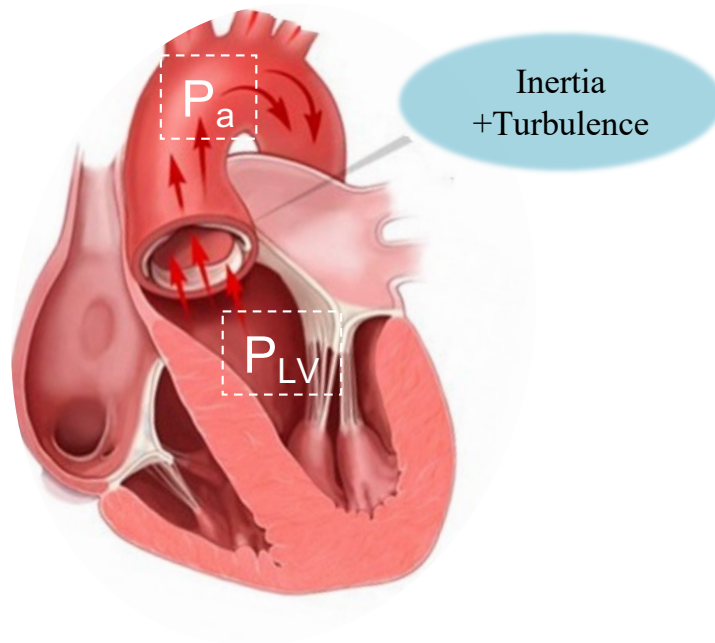


Heart model



Varying elastance model

Aortic valve effects



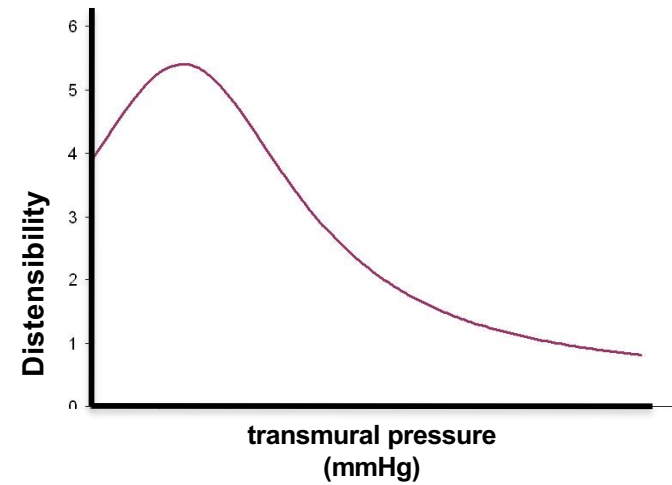
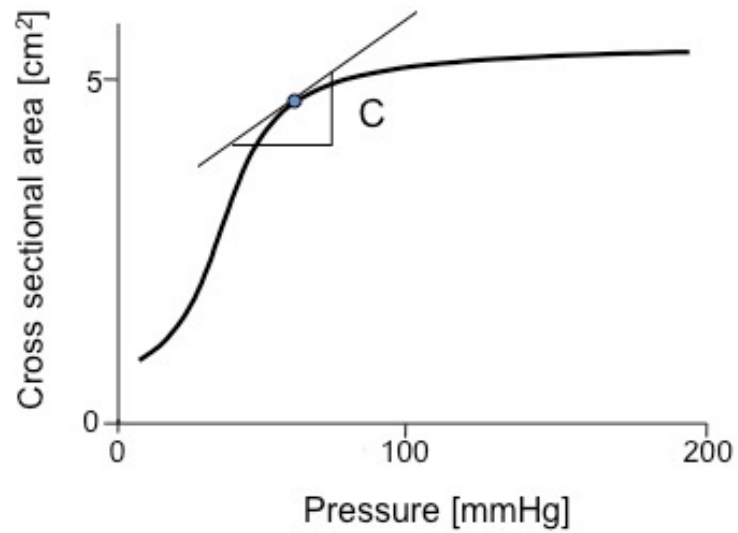
Inclusion of phenomena of inertia and turbulence occurring at the aortic valve

Improve wave shape quality

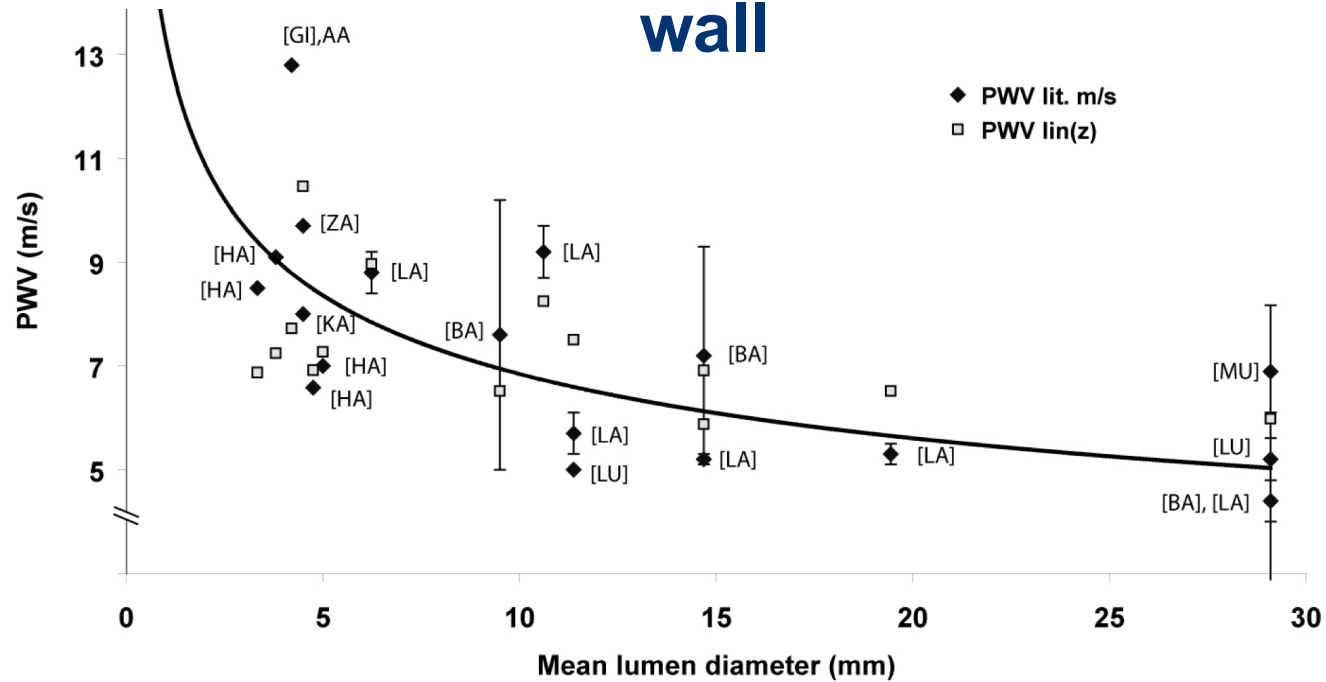
$$\Delta p = P_{lv} - P_a = BQ|Q| + L \frac{dQ}{dt}$$

Mynard JP et al. Int J Numer Methods Biomed Eng 2012;28:626–41.

Viscoelastic properties of the arterial wall



Viscoelastic properties of the arterial wall



Nonlinear Viscoelastic behavior

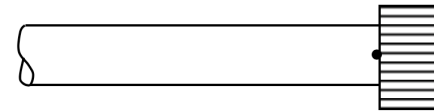
$$A^v(t) = \int_0^\infty \dot{J}(\tau) A^e(P(t - \tau)) d\tau$$

$$\dot{J}(\tau) = \tilde{a} \frac{e^{-\tau/\tau_2} - e^{-\tau/\tau_1}}{\tau}$$

Holenstein, Niederer et al. 1980

Downstream boundary conditions

- Specify either:
 1. pressure waveform
 2. flow waveform or
 3. the terminal (distal) impedance (Z_T)
- The terminal impedance model of vasculature beyond the terminal point and characterizes the relation between pressure and flow at the terminal point.



$$Z_T = \frac{P}{Q}$$

Boundary condition at bifurcation

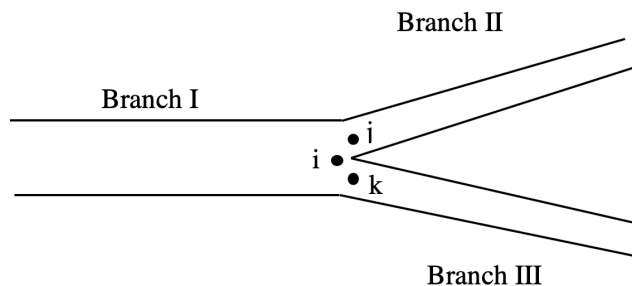
Apply a simple continuity in pressure and flow, such as described below.

Indices:

i = last node of mother branch I

j = first node of daughter branch II

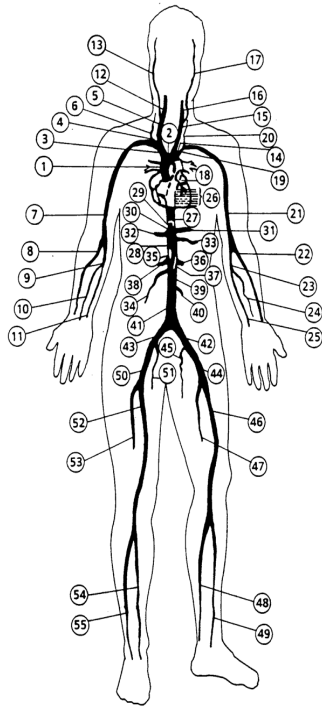
k = first node of daughter branch III



$$p_i = p_j = p_k$$

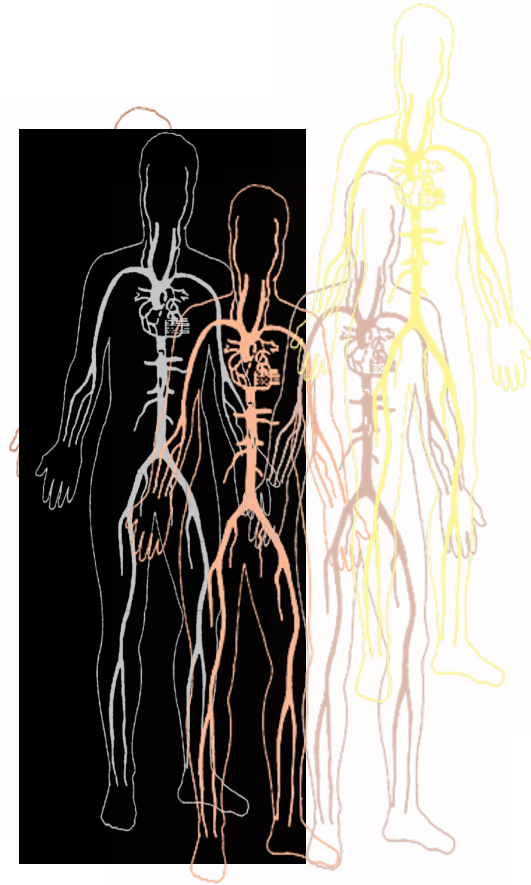
$$Q_i = Q_j + Q_k$$

Validation strategy



Generic 1-D model

Qualitative
assessment of
pressure and
flow wave
agreement



Group of young volunteers

In vivo measurements

In vivo measurements

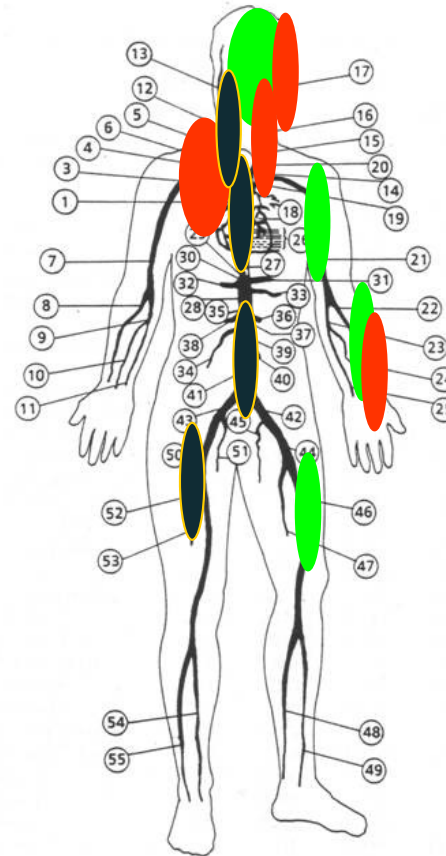
Flow rate waveforms

MRI

Ultrasound Doppler
(Supine position, basal conditions)

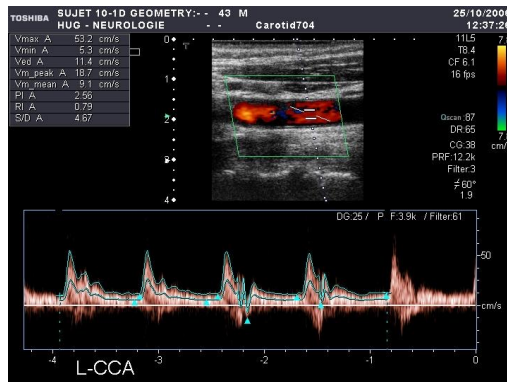
Pressure waveforms

applanation Tonometry

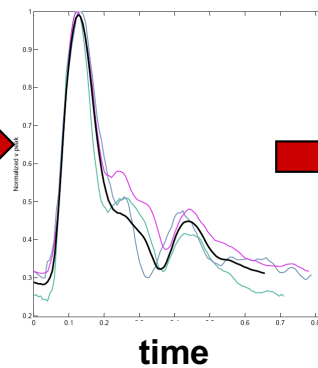


Flow from Doppler velocity

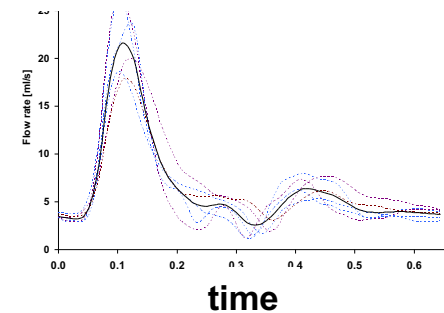
e.g. CCA (15-30 y)
yrs. 26 ± 4



peak velocities (n=6)



flow rates (n=6)

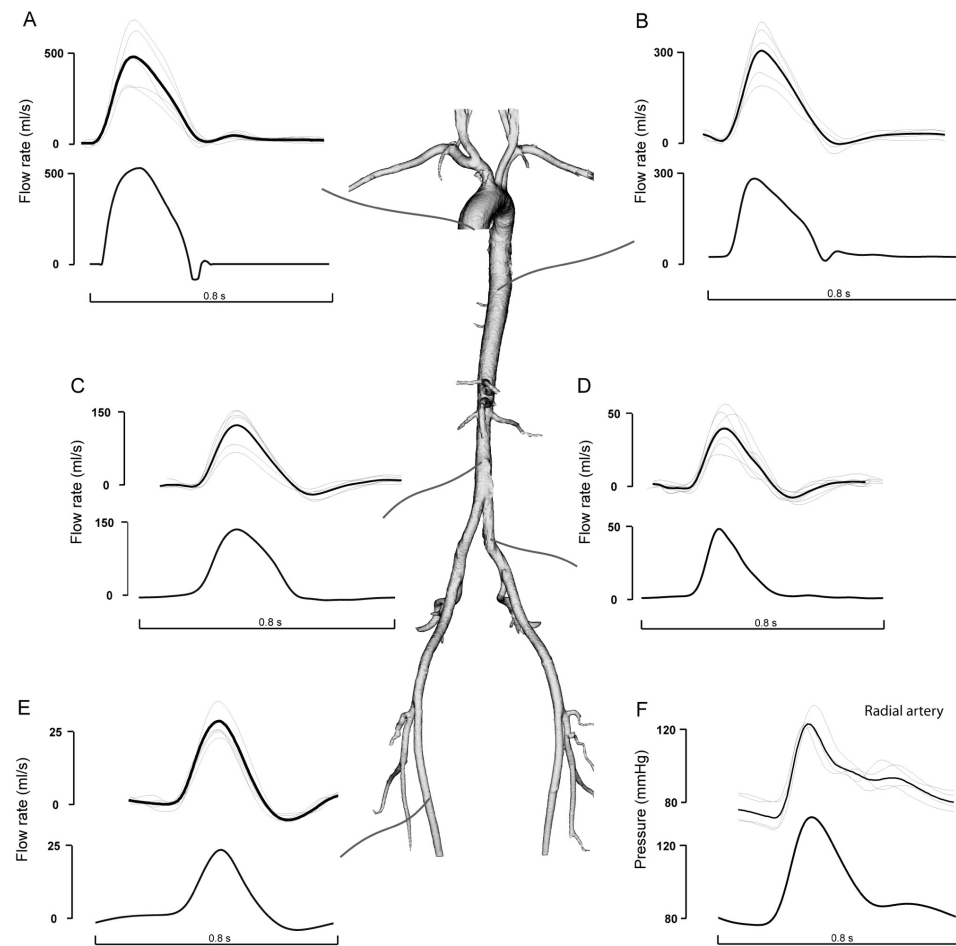


Extract **peak velocity**
along each cycle

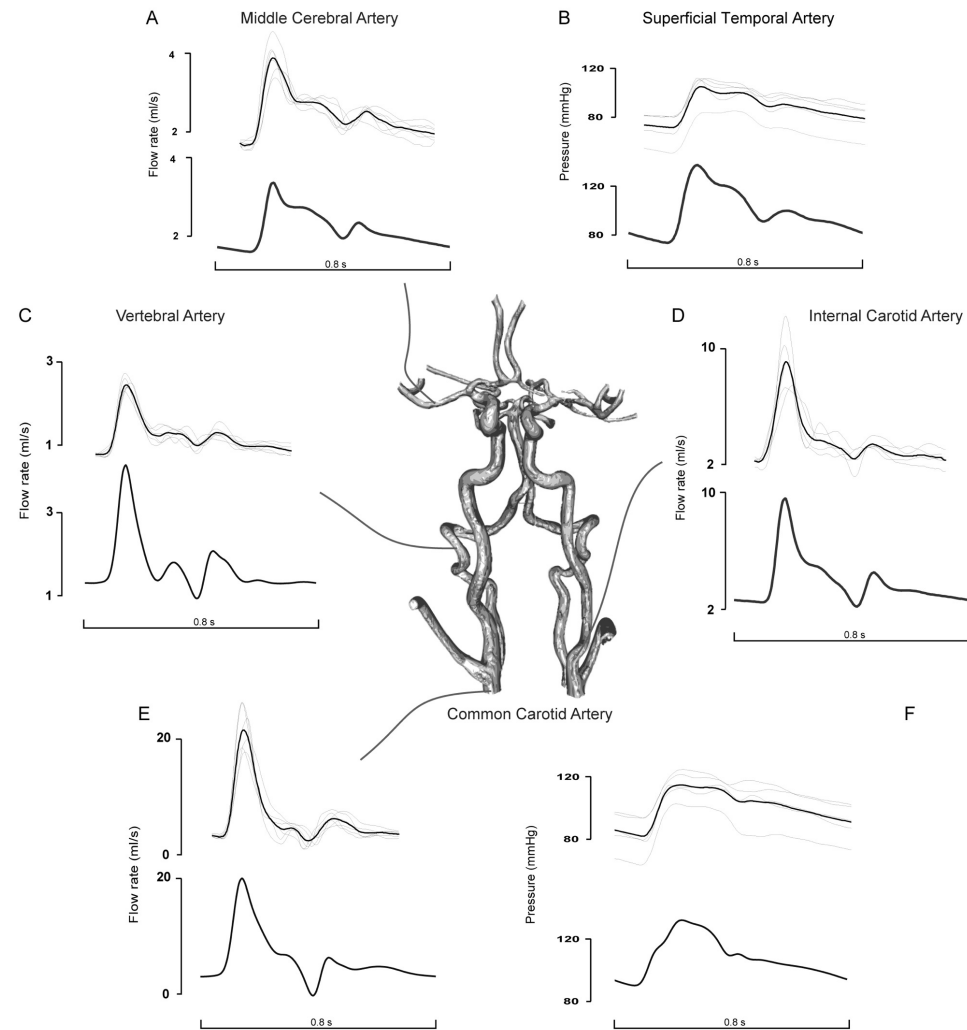
use of Womersley's
theory of pulsatile flow in rigid &
cylindrical vessel

assumption:
known diameter

Flow & pressure waveforms in aorta & limbs

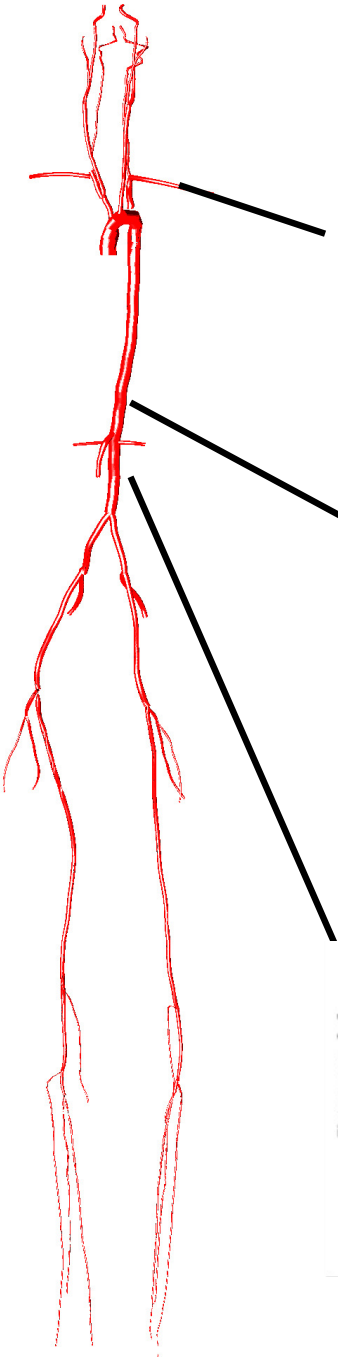


Flow & pressure waveforms in cerebral arteries

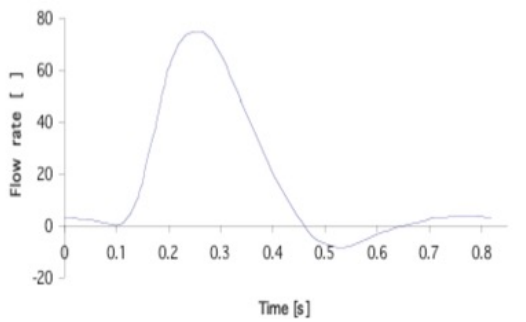
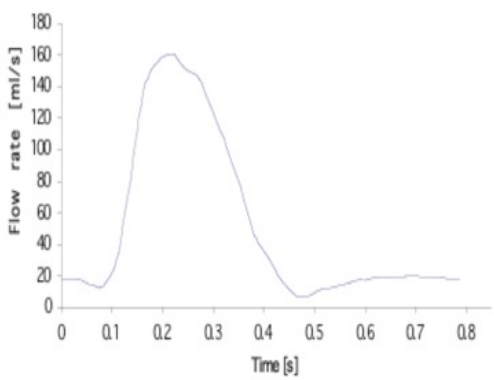
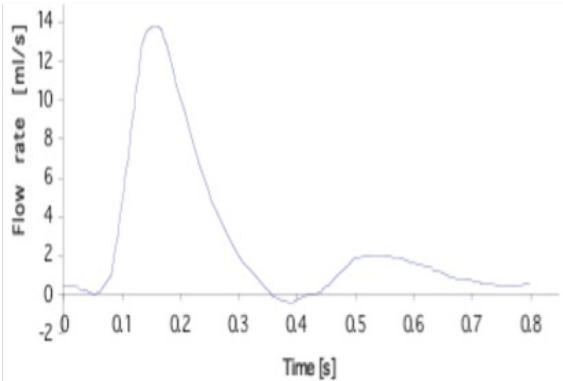


Reymond et al., AJP,
2009

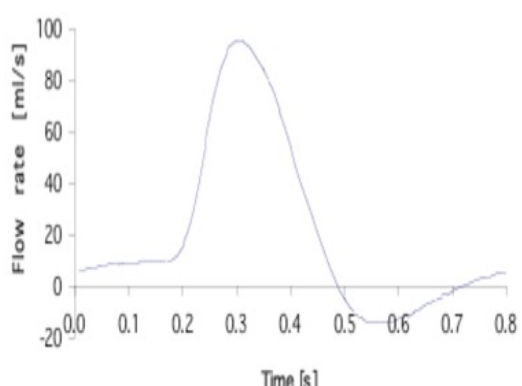
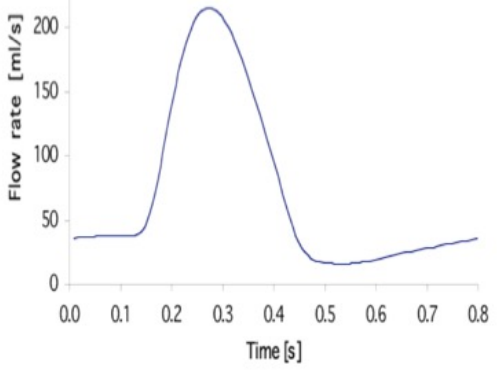
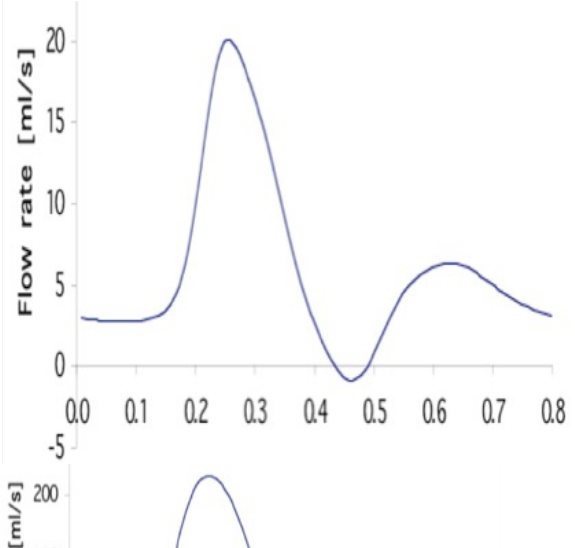
Flowrate waveforms
Main systemic arteries



MRI

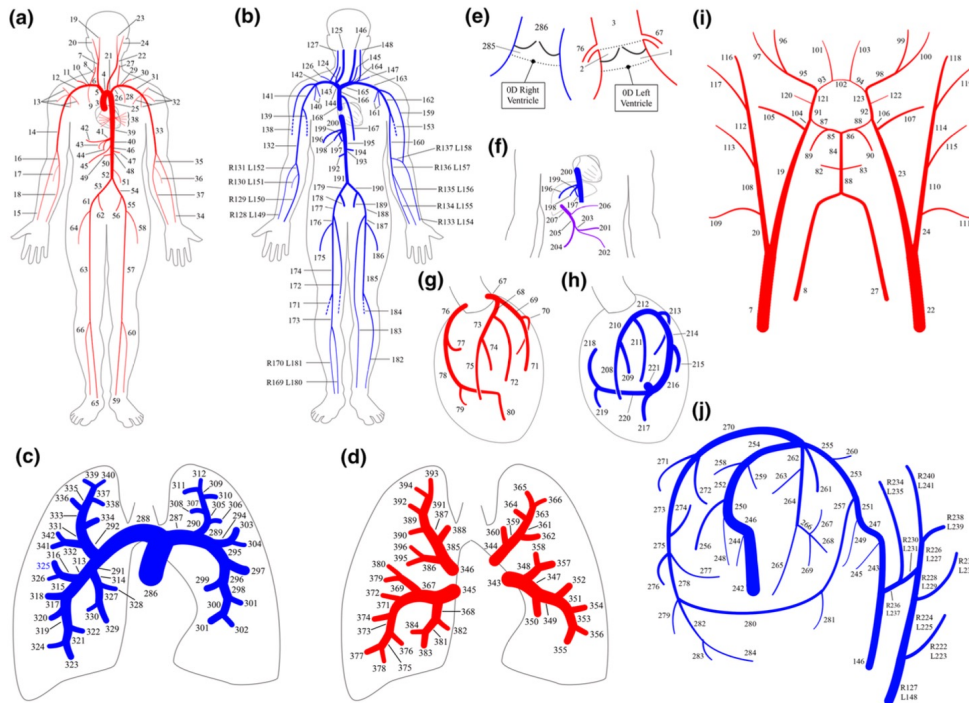


Model

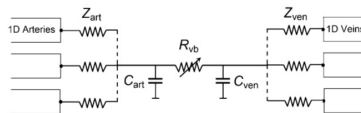


1-D model of the entire circulation

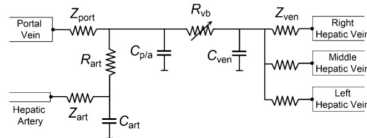
J. P. Mynard and J. J. Smolich,
 “One-Dimensional Haemodynamic Modeling and Wave Dynamics in the Entire Adult Circulation”,
 Annals of Biomedical Engineering, Vol. 43, No. 6, June 2015 pp. 1443–1460



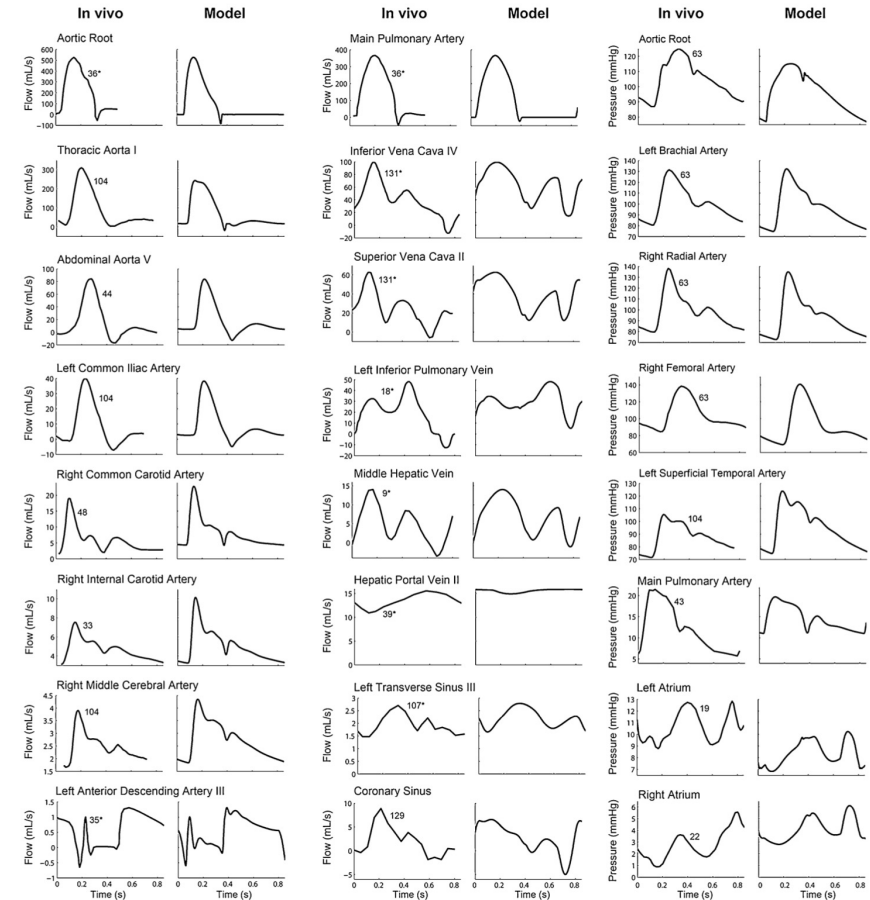
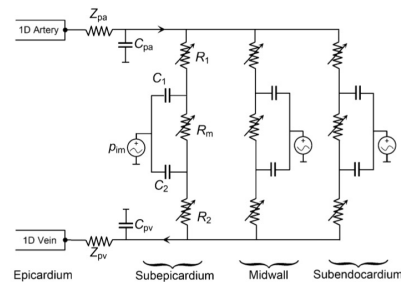
(a) Generic Vascular Bed



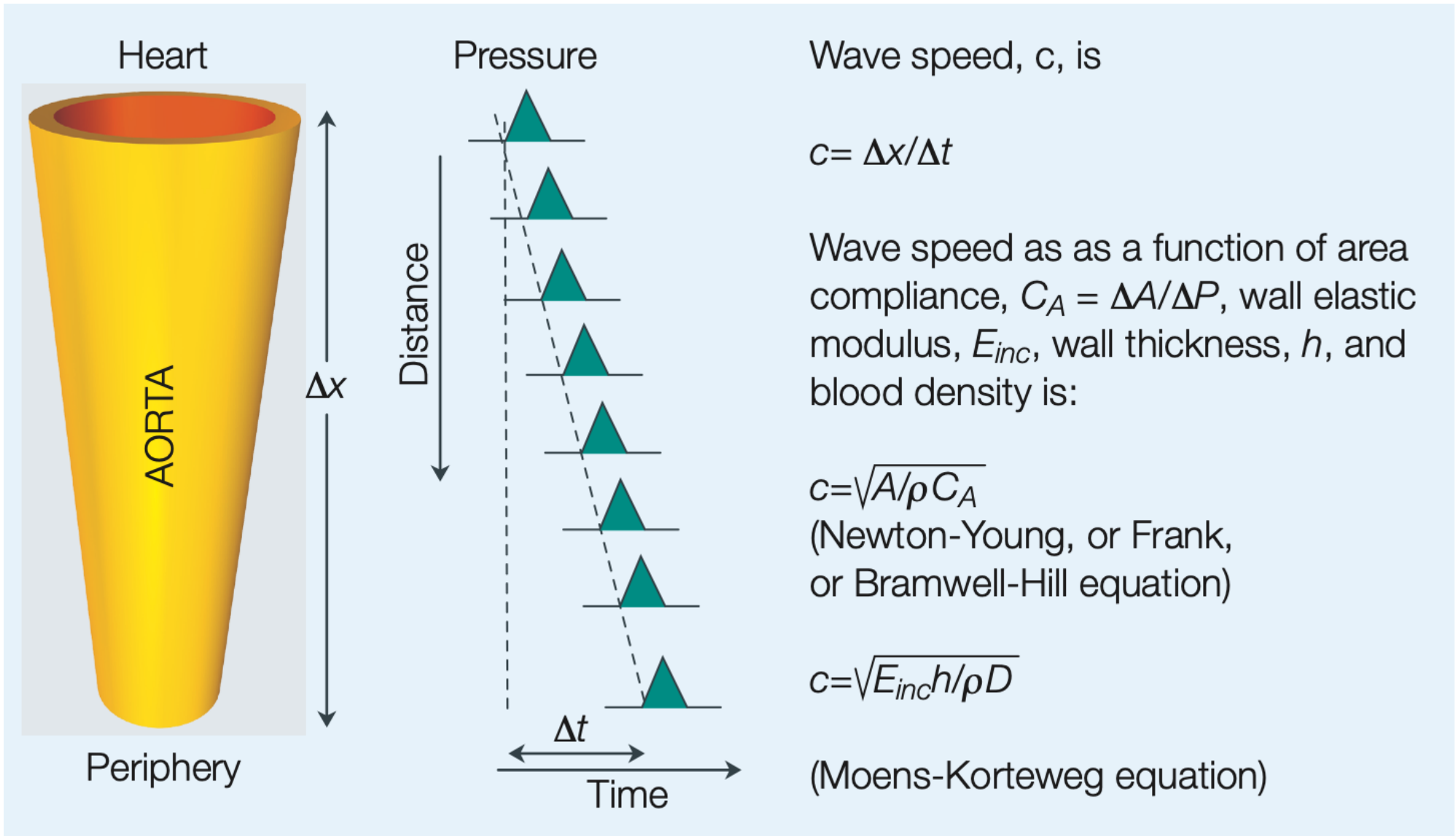
(b) Hepatic Vascular Bed



(c) Coronary Vascular Bed



Wave travel & wave speed



Relation of wave speed to arterial compliance

Continuity: $\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$

Momentum: $\frac{\partial Q}{\partial t} + \frac{A}{\rho} \frac{\partial P}{\partial x} + \pi \frac{2\pi R}{\rho} = 0$

If we neglect friction: $\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0$
 $\frac{\partial Q}{\partial t} + \frac{A}{\rho} \frac{\partial P}{\partial x} = 0$

The area compliance is defined as: $C_A = \frac{\partial A}{\partial P}$

$$\frac{\partial A}{\partial t} = \frac{\partial A}{\partial P} \frac{\partial P}{\partial t} = C_A \frac{\partial P}{\partial t}$$

The 1D flow equations can be rewritten as:

$$\left. \begin{aligned} C_A \frac{\partial P}{\partial t} + \frac{\partial Q}{\partial x} &= 0 \\ \frac{\partial Q}{\partial t} + \frac{A}{\rho} \frac{\partial P}{\partial x} &= 0 \end{aligned} \right\} \Rightarrow \begin{aligned} C_A \frac{\partial^2 P}{\partial t^2} + \frac{\partial^2 Q}{\partial x \partial t} &= 0 \\ \frac{\partial^2 Q}{\partial x \partial t} + \frac{A}{\rho} \frac{\partial^2 P}{\partial x^2} &= 0 \end{aligned} \Rightarrow \underline{\underline{\frac{\partial^2 P}{\partial t^2} = \frac{A}{\rho C_A} \frac{\partial^2 P}{\partial x^2}}} \quad \text{Wave equation!}$$

where the wave speed is: $c^2 = \frac{A}{\rho C_A}$

or $c = \sqrt{\frac{A}{\rho} \frac{1}{C_A}}$

Newton-Young
or
Frank equation
or
Bramwell-Hill

The wave speed, c , is often also called Pulse Wave Velocity, PWV

Relation of wave speed to arterial wall elasticity

Bramwell-Hill relation: $c = \sqrt{\frac{A}{\rho} \frac{1}{C_A}}$ (1)

Area compliance: $C_A = \frac{\partial A}{\partial P} = \frac{\partial(\pi R^2)}{\partial P} = 2\pi R \frac{\partial R}{\partial P}$ (2)

$$(1) \wedge (2) \Rightarrow c = \sqrt{\frac{\pi R^2}{\rho} \cdot \frac{1}{2\pi R} \frac{\partial P}{\partial R}} = \sqrt{\frac{R}{2\rho} \frac{\partial P}{\partial R}} \quad (3)$$

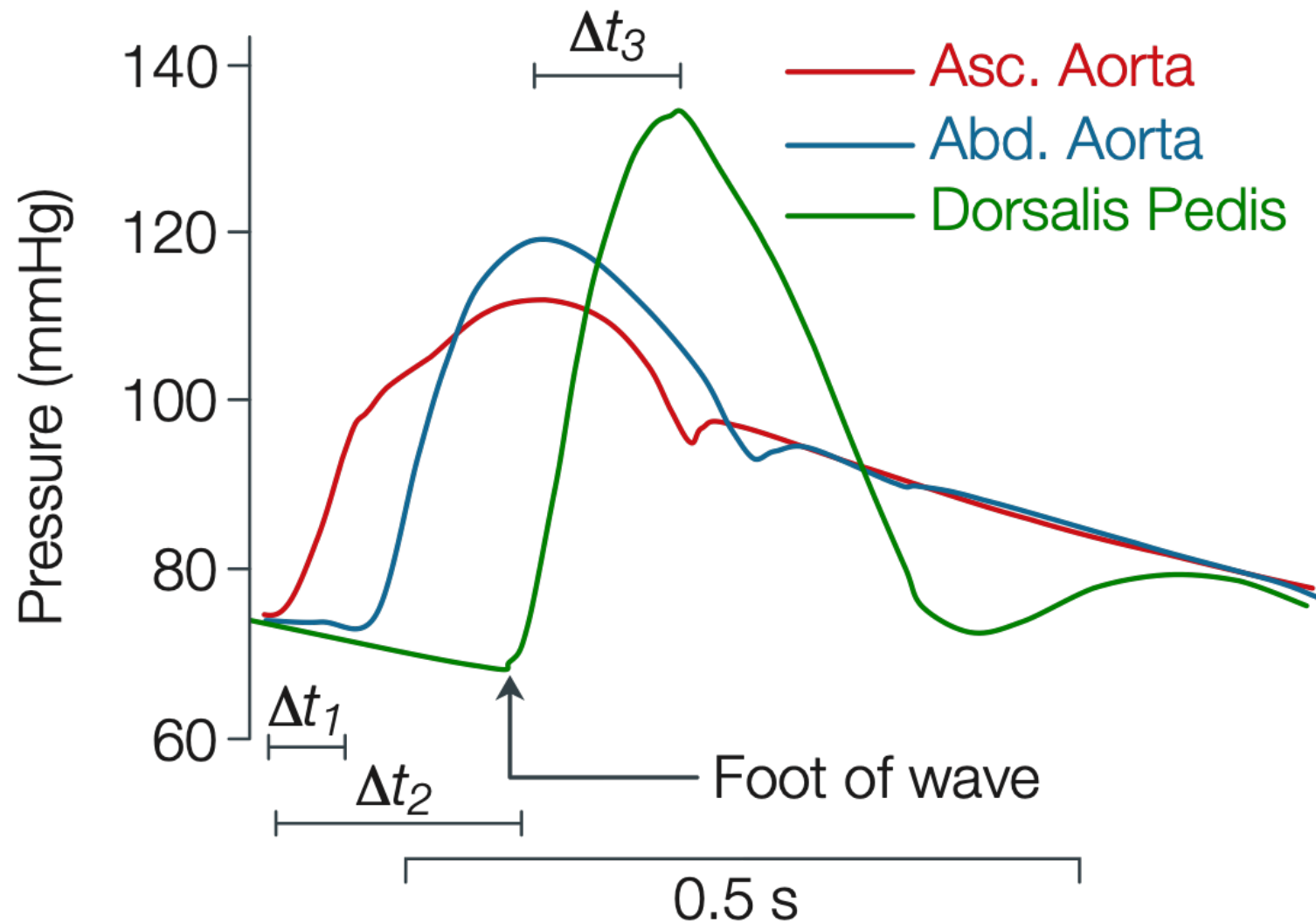
The incremental elastic modulus of the arterial wall is given by:

$$E_{inc} = \frac{R^2}{h} \frac{\partial P}{\partial R} \Rightarrow \frac{\partial R}{\partial P} = \frac{R^2}{h} \cdot \frac{1}{E_{inc}} \quad (4)$$

$$(3) \wedge (4) \Rightarrow c = \sqrt{\frac{R}{2\rho} \cdot \frac{h E_{inc}}{R^2}} = \sqrt{\frac{E_{inc} \cdot h}{\rho \cdot 2R}}$$

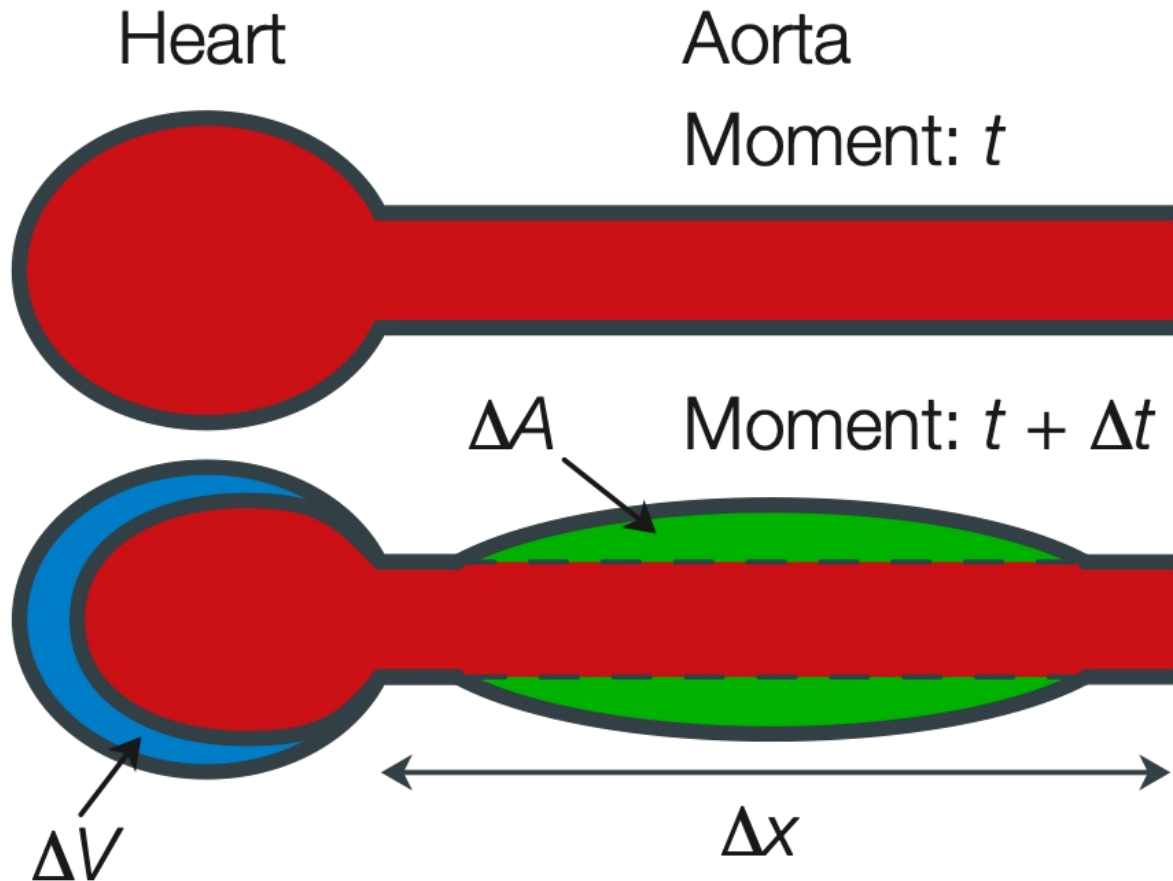
or $c = \sqrt{\frac{E_{inc} \cdot h}{\rho D}}$ Moens-Korteweg relation

Foot-to-foot method to derive wave speed



$c = \frac{\Delta x}{\Delta t}$ is valid only at the foot of the wave, were the wave is unidirectional (no reflected waves)

Derive PWV from the relation of area to flow



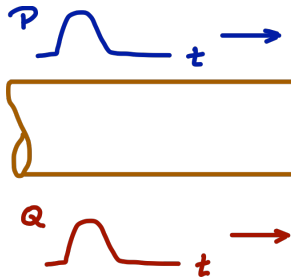
$$\Delta V = \Delta A \cdot \Delta x$$

$$\frac{\Delta V}{\Delta t} = \Delta A \cdot \frac{\Delta x}{\Delta t}$$

$$\Delta Q = \Delta A \cdot c$$

$$c = \frac{\Delta Q}{\Delta A}$$

Characteristic impedance, Z_c



$$Z_c = \frac{P}{Q}$$

Definition: Characteristic impedance, Z_c , is the ratio of pulsatile pressure to flow in case of unidirectional waves (no reflections)

Relation of Z_c to geometry and elastic properties

Let $P(t)$ be a single harmonic wave $P(t) = P_0 e^{i\omega(t - \frac{x}{c})}$

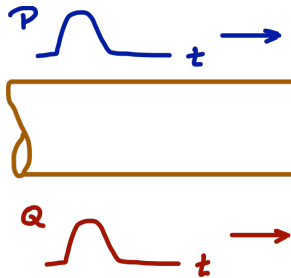
$$\text{Continuity: } \frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0 \Rightarrow \underbrace{\frac{\partial A}{\partial P}}_{C_A} \frac{\partial P}{\partial t} + \underbrace{\frac{\partial Q}{\partial P}}_{\frac{1}{Z_c}} \frac{\partial P}{\partial x} = 0 \Rightarrow C_A \frac{\partial P}{\partial t} + \frac{1}{Z_c} \frac{\partial P}{\partial x} = 0$$

$$\Rightarrow C_A \cdot i\omega P_0 e^{i\omega(t - \frac{x}{c})} + \frac{1}{Z_c} \cdot \left(-\frac{i\omega}{c}\right) \cdot P_0 e^{i\omega(t - \frac{x}{c})} = 0 \Rightarrow C_A - \frac{1}{Z_c c} = 0$$

$$\Rightarrow Z_c = \frac{1}{C C_A} = \frac{1}{\sqrt{\frac{A}{\rho} \cdot \frac{1}{C_A}} \cdot C_A} = \underline{\underline{\sqrt{\frac{\rho}{A} \cdot \frac{1}{C_A}}}}$$

$$\text{Since } c = \sqrt{\frac{A}{\rho} \cdot \frac{1}{C_A}} \Rightarrow \frac{Z_c}{c} = \frac{\rho}{A} \Rightarrow \underline{\underline{Z_c = \frac{\rho c}{A}}}$$

Methods to derive the characteristic impedance, Z_c



$$Z_c = \frac{P}{Q}$$

Definition: Characteristic impedance, Z_c , is the ratio of pulsatile pressure to flow in case of unidirectional waves (no reflections)

1) Direct (definition)

A. Frequency domain: $Z_c = \frac{P(\omega)}{Q(\omega)}$

B. Time domain: $Z_c = \frac{\Delta P}{\Delta Q}$



Attention: good only for unidirectional waves

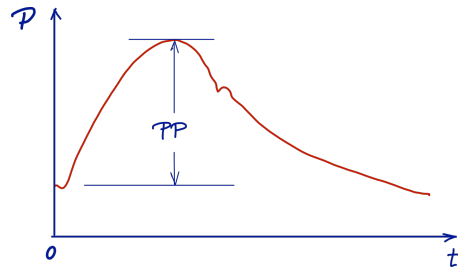
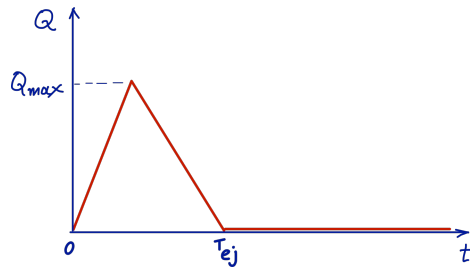
3) From area compliance and geometry:

$$Z_c = \sqrt{\frac{\rho}{A} \frac{1}{C_A}}$$

4) From wave speed and geometry:

$$Z_c = \frac{\rho c}{A}$$

Example: estimate aortic pulse pressure from cardiac output



Assume that aortic flow can be approximated by a triangular wave during the ejection phase, as shown in the Figure. Estimate the **aortic pulse pressure** assuming that:

1. Cardiac output is 5.4 lit/min
2. Heart rate is 72 beat/min
3. Duration of the ejection phase is $T_{ej} = 250$ ms
4. Ascending aortic diameter is 2.6 cm
5. The wave speed in the ascending aorta is 4.5 m/s

a) The stroke volume is: $Q = SV \cdot HR \Rightarrow SV = \frac{Q}{HR} = \frac{5.4 \text{ lit/min}}{72 \text{ beats/min}} = 0.075 \text{ lit} = \underline{\underline{75 \text{ ml}}}$

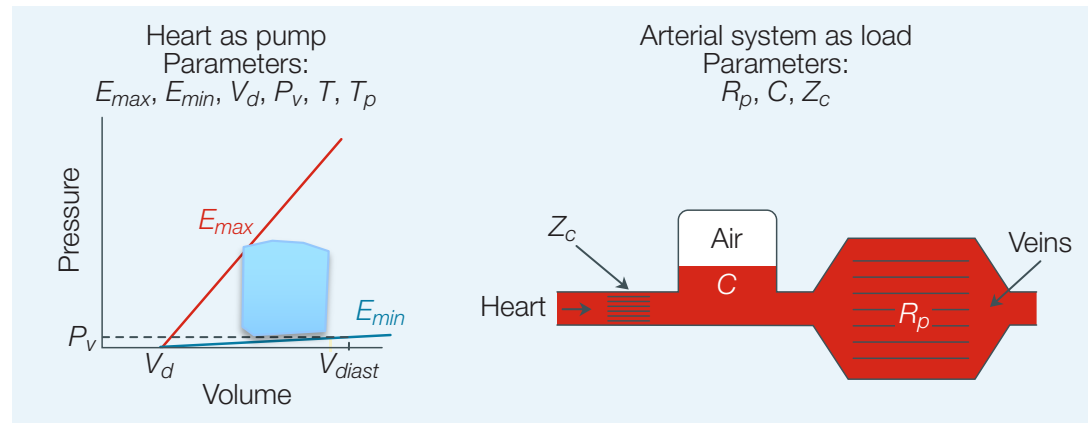
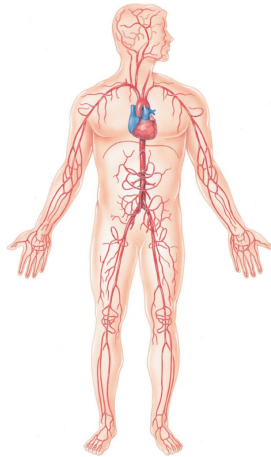
b) $SV = \text{area under the systolic flow curve} \Rightarrow SV = \frac{1}{2} Q_{max} \cdot T_{ej}$
 $\Rightarrow Q_{max} = \frac{2 \cdot SV}{T_{ej}} = \frac{2 \cdot 75 \text{ ml}}{0.250 \text{ s}} = \underline{\underline{600 \text{ ml/s}}}$

c) $Z_c = \frac{\rho c}{A} = \frac{1050 \text{ kg/m}^3 \cdot 4.5 \text{ m/s}}{\pi \cdot 0.013^2 \text{ m}^2} = \underline{\underline{8.9 \times 10^6 \frac{\text{kg}}{\text{m}^4 \cdot \text{s}}}}$

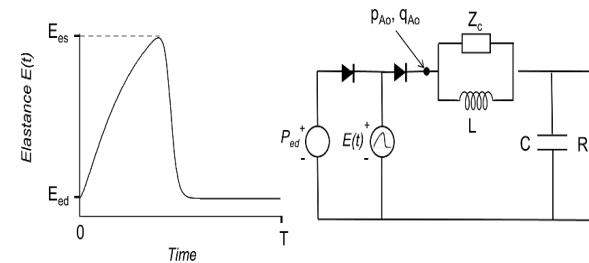
d) Assuming no wave reflections in early systole: $Z_c = \frac{\Delta P}{\Delta Q} = \frac{PP}{Q_{max}}$

$\Rightarrow PP = Z_c \cdot Q_{max} = 8.9 \times 10^6 \frac{\text{kg}}{\text{m}^4 \cdot \text{s}} \times 600 \times 10^{-6} \frac{\text{m}^3}{\text{s}} = 5340 \frac{\text{N}}{\text{m}^2} \Rightarrow \underline{\underline{PP = 40 \text{ mmHg}}}$

Global approach to heart-arterial system interaction

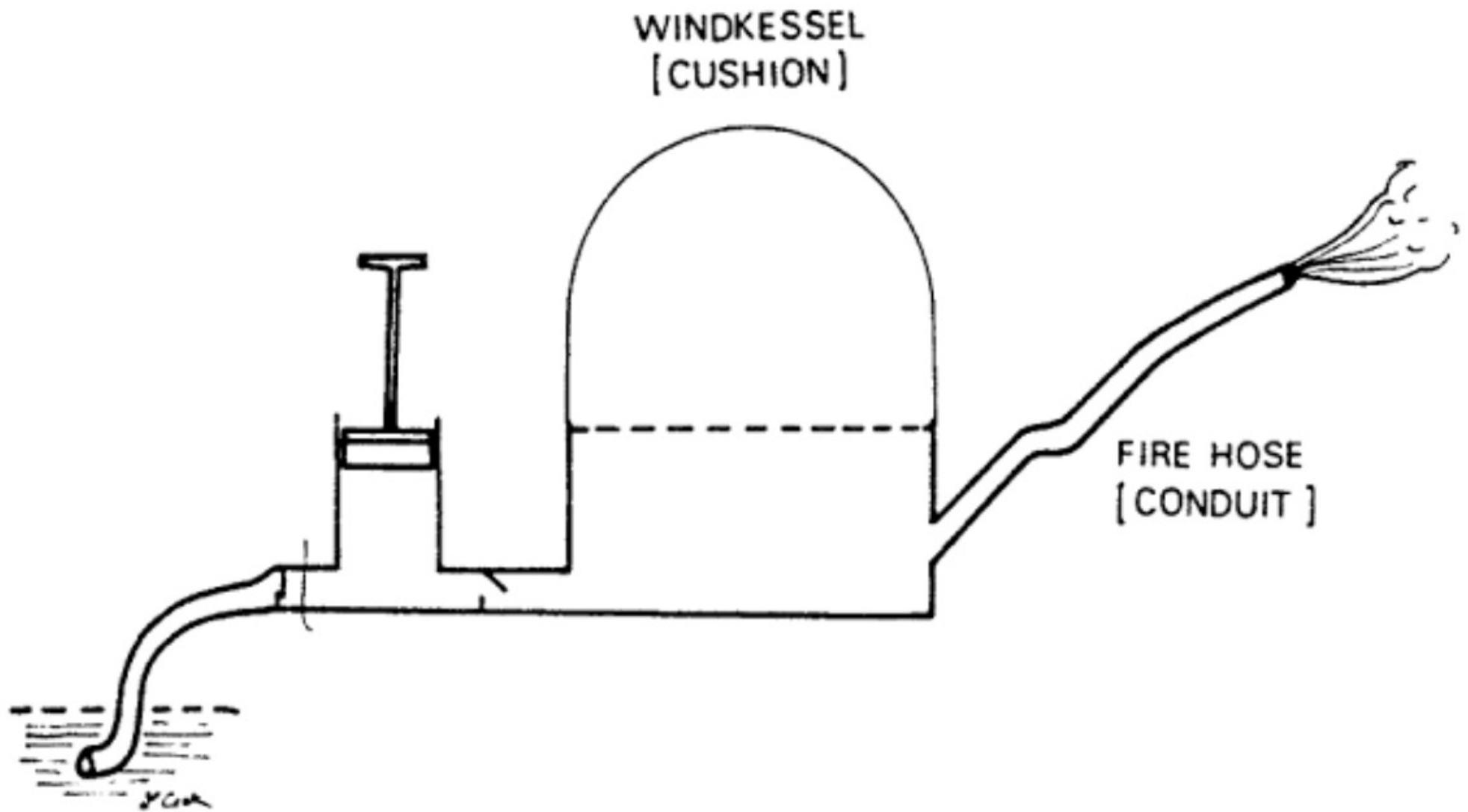


- ❑ 4-element Windkessel model of the arterial tree coupled with a simple heart (elastance) model by Maksuti et al. (2016)
- ❑ Quite accurate, BUT does not capture wave transmission effects

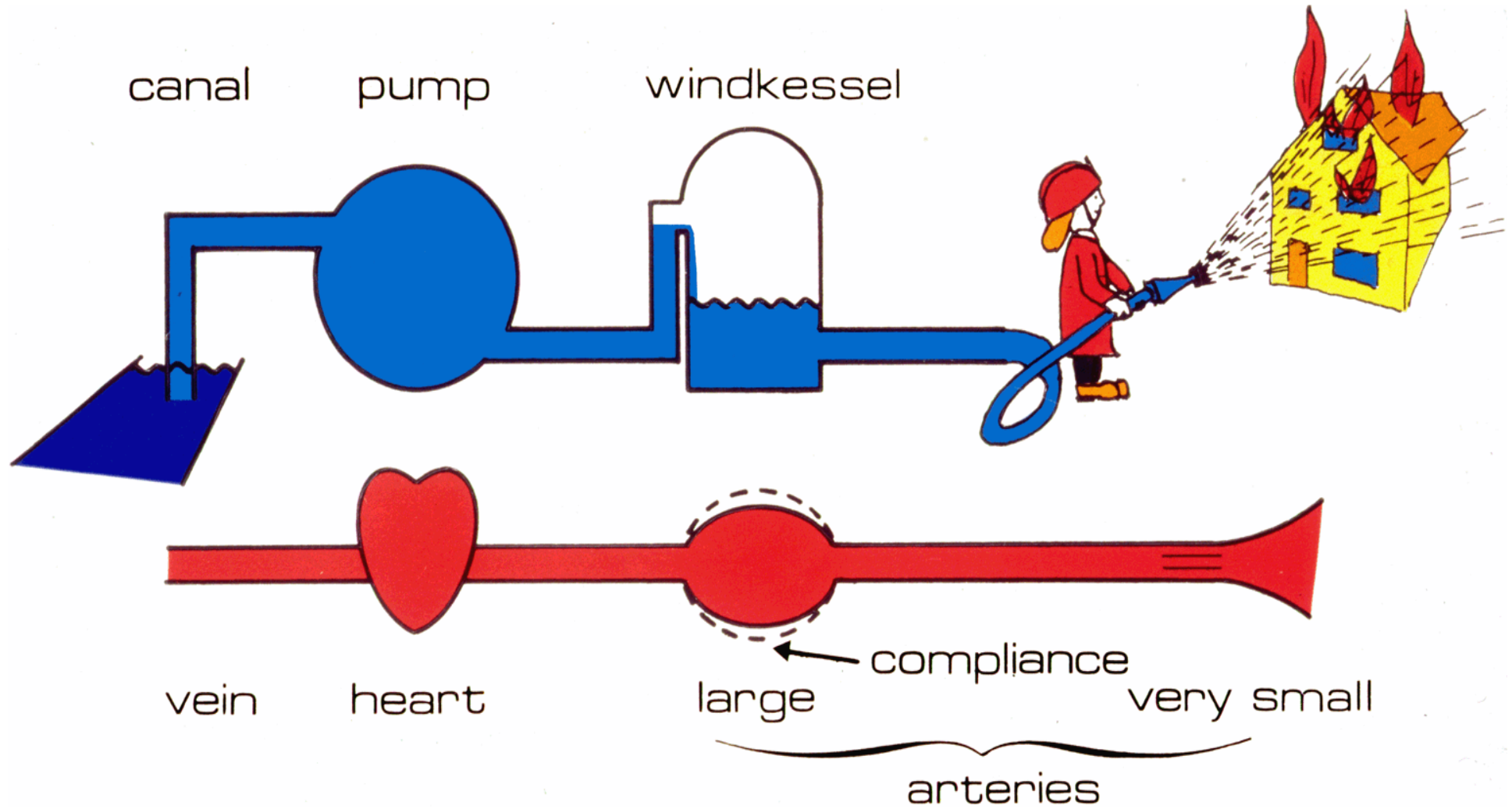


Source: Maksuti et al. Contribution of the Arterial System and the Heart to Blood Pressure during Normal Aging - A Simulation Study. PloS One 2016;11:e0157493.

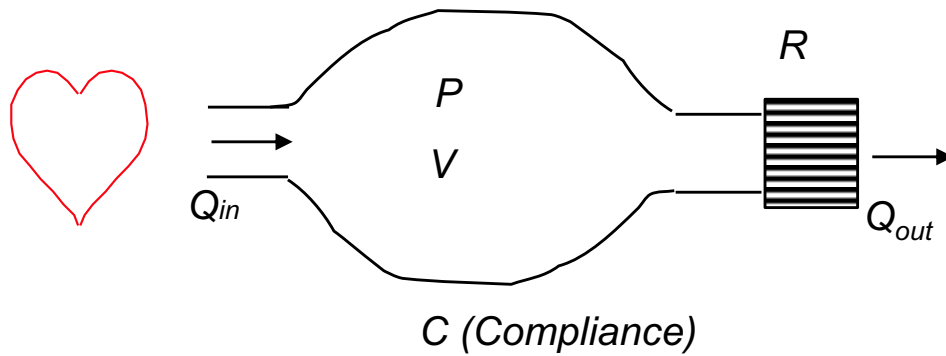
Fire Engine as model



Fire Engine as model



2-element windkessel model (Otto Frank, 1899)

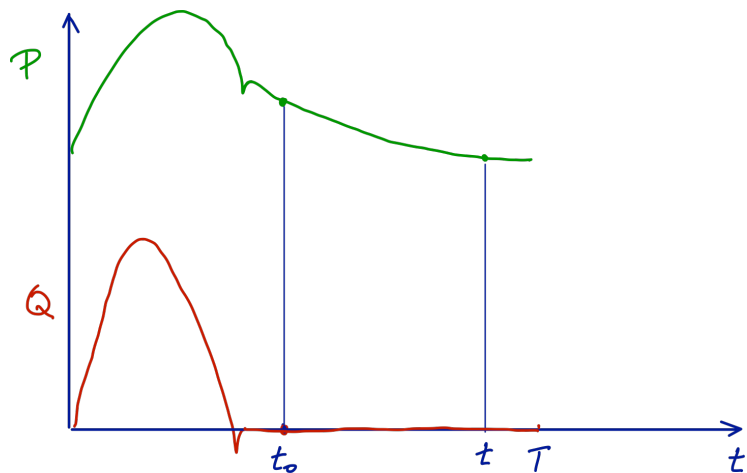


$$R = \frac{P}{Q_{out}}$$

$$C = \frac{dV}{dP}$$

Continuity: $Q_{in} - Q_{out} = \frac{dV}{dt} = \underbrace{\frac{dV}{dP}}_{\text{Volume Compliance}} \frac{dP}{dt} = C \frac{dP}{dt} \Rightarrow \underline{\underline{C \frac{dP}{dt} + \frac{P}{R} = Q_{in}}} \quad (\text{time domain})$

During diastole: $Q_{in} = 0$

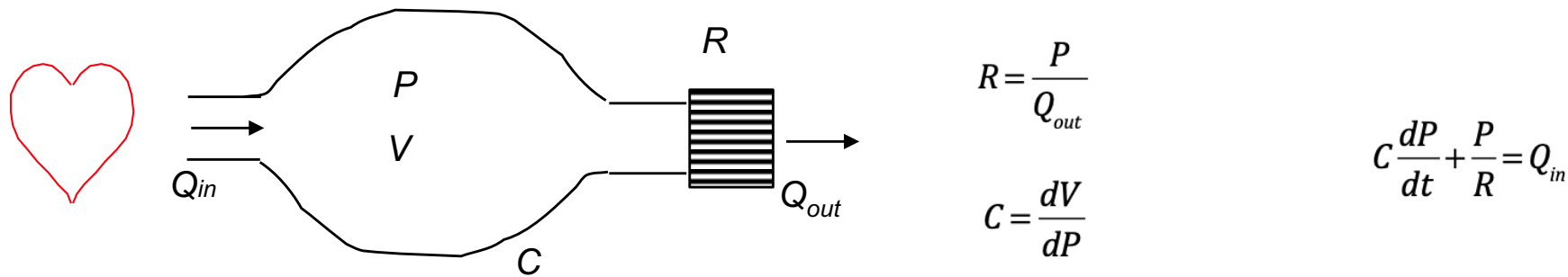


$$\Rightarrow C \frac{dP}{dt} + \frac{P}{R} = 0 \Rightarrow \frac{dP}{dt} = - \frac{P}{RC}$$

$$\Rightarrow \frac{dP}{P} = - \frac{dt}{RC} \Rightarrow \int_{P_0}^P \frac{dP}{P} = \int_{t_0}^t \frac{dt}{RC}$$

$$\Rightarrow \ln \frac{P}{P_0} = - \frac{t - t_0}{RC} \Rightarrow P(t) = P_0 e^{-\frac{t - t_0}{RC}}$$

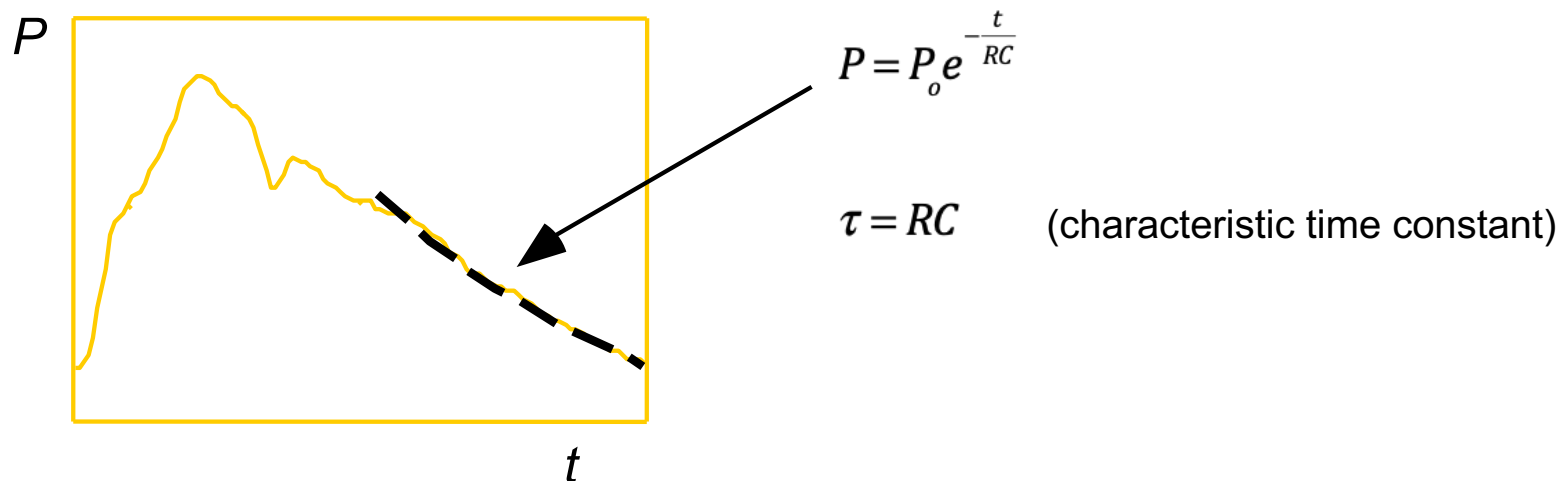
2-element windkessel model (Otto Frank, 1899)



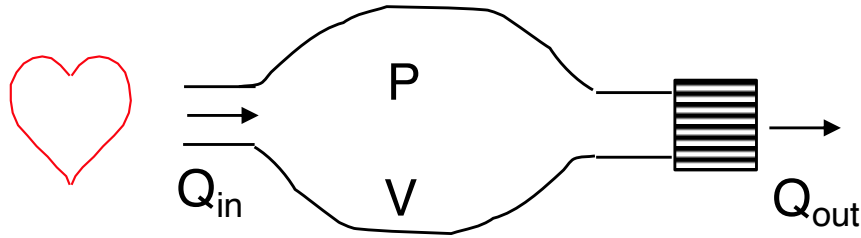
Method to derive total arterial compliance:

- Fit an exponential decay function to the diastolic part of the curve
- The characteristic time constant $\tau = RC$

- Calculate total peripheral resistance as mean pressure over mean flow: $R = \frac{\bar{P}}{\bar{Q}}$
- Calculate total arterial compliance as: $C = \frac{\tau}{R}$

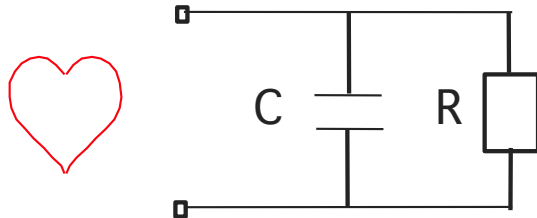


Electrical analog of the 2-element windkessel model



Time domain: $C \frac{dP}{dt} + \frac{P}{R} = Q$

Electrical analog



Total impedance, Z_{in} is an addition in parallel of R and C

$$\frac{1}{Z_{in}} = \frac{1}{R} + \frac{1}{\frac{1}{j\omega C}} = \frac{1}{R} + j\omega C = \frac{1 + j\omega RC}{R}$$

$$\Rightarrow Z_{in} = \frac{R}{1 + j\omega RC} \quad (\text{frequency domain})$$

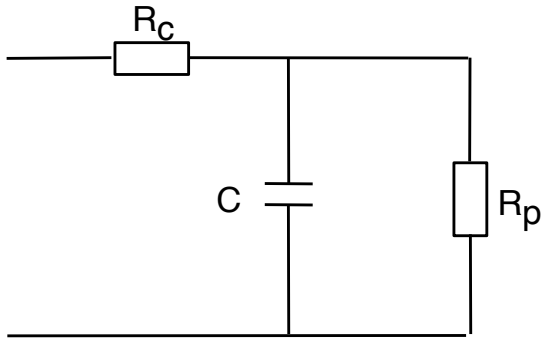
Check if time domain description is the same as the frequency domain one:

$$Z_{in} = \frac{P}{Q} = \frac{R}{1 + j\omega RC} \quad (\text{frequency domain}) \Rightarrow P + j\omega RC P = R \cdot Q$$

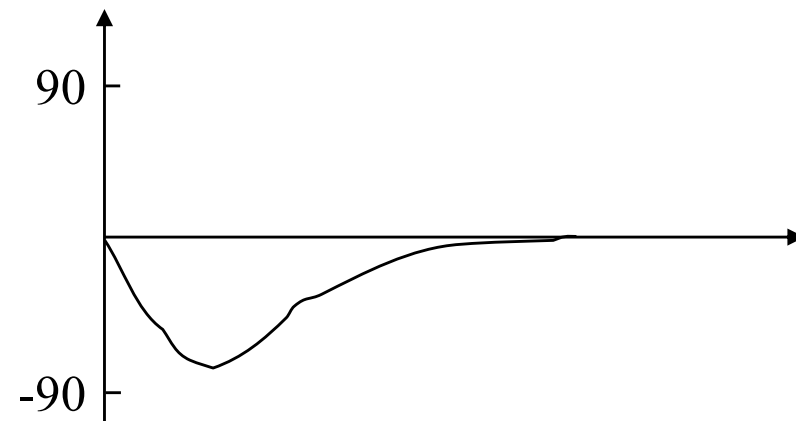
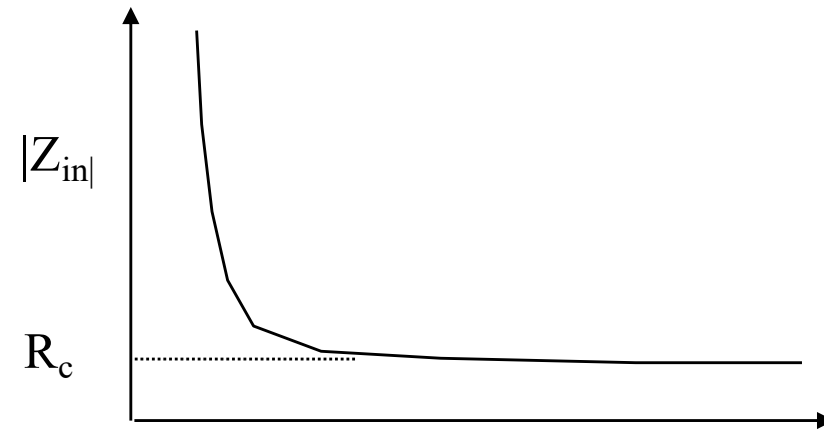
$$\left. \begin{array}{l} P + j\omega RC P = R \cdot Q \\ \text{Note: } j\omega P = \frac{dP}{dt} \end{array} \right\} \Rightarrow C \frac{dP}{dt} + \frac{P}{R} = Q \quad (\text{time domain})$$

Three-element windkessel

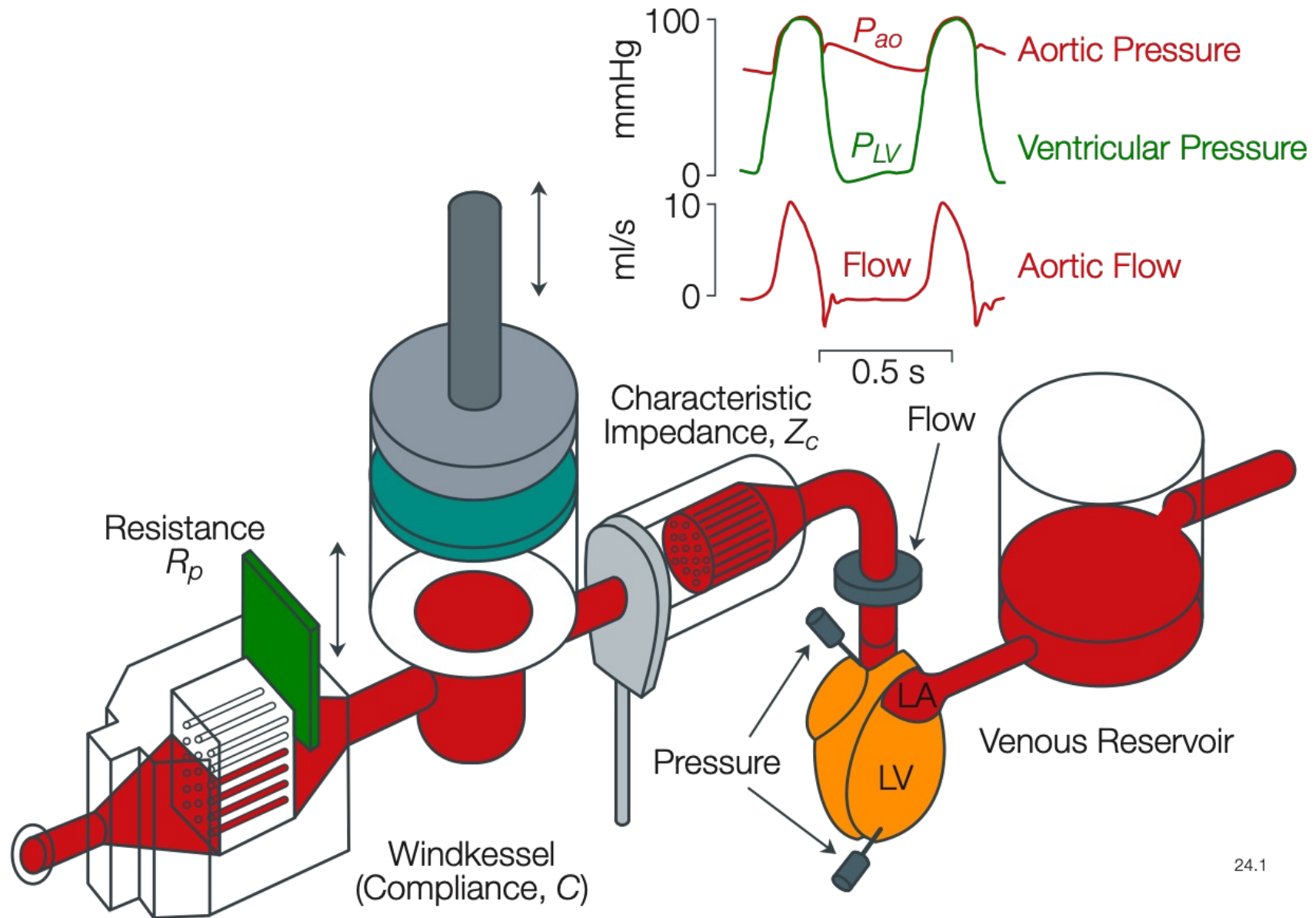
Three-element windkessel



$$Z_{in} = R_c + \frac{R_p}{1 + j\omega R_p C}$$

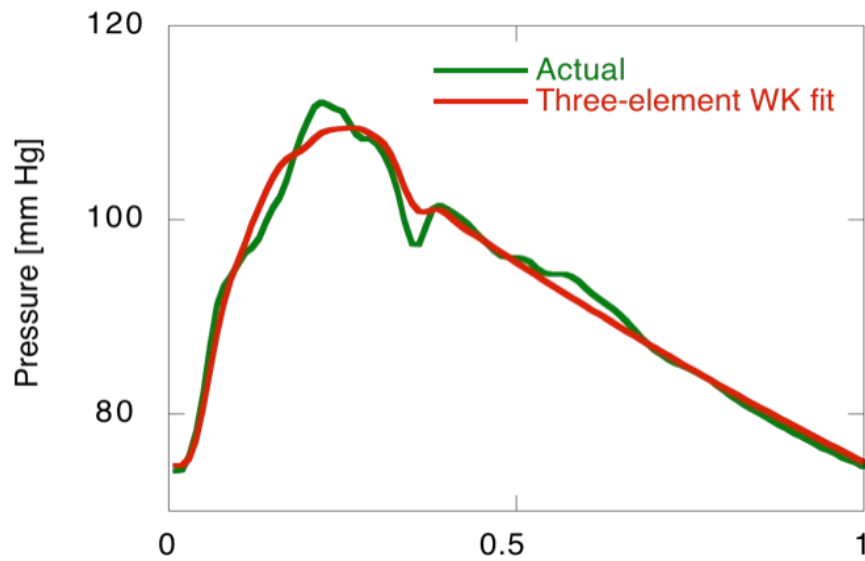


3-element windkessel

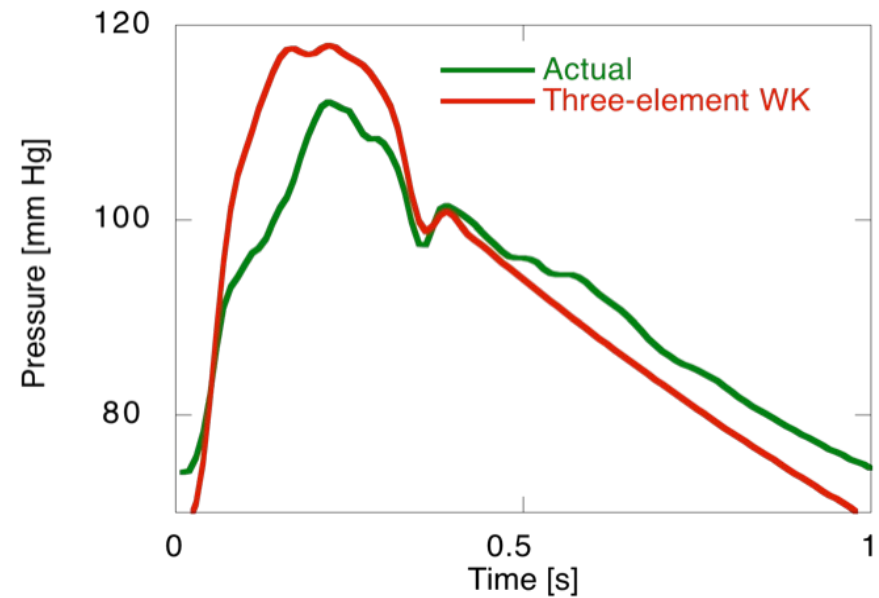


The problem with the 3-element WK

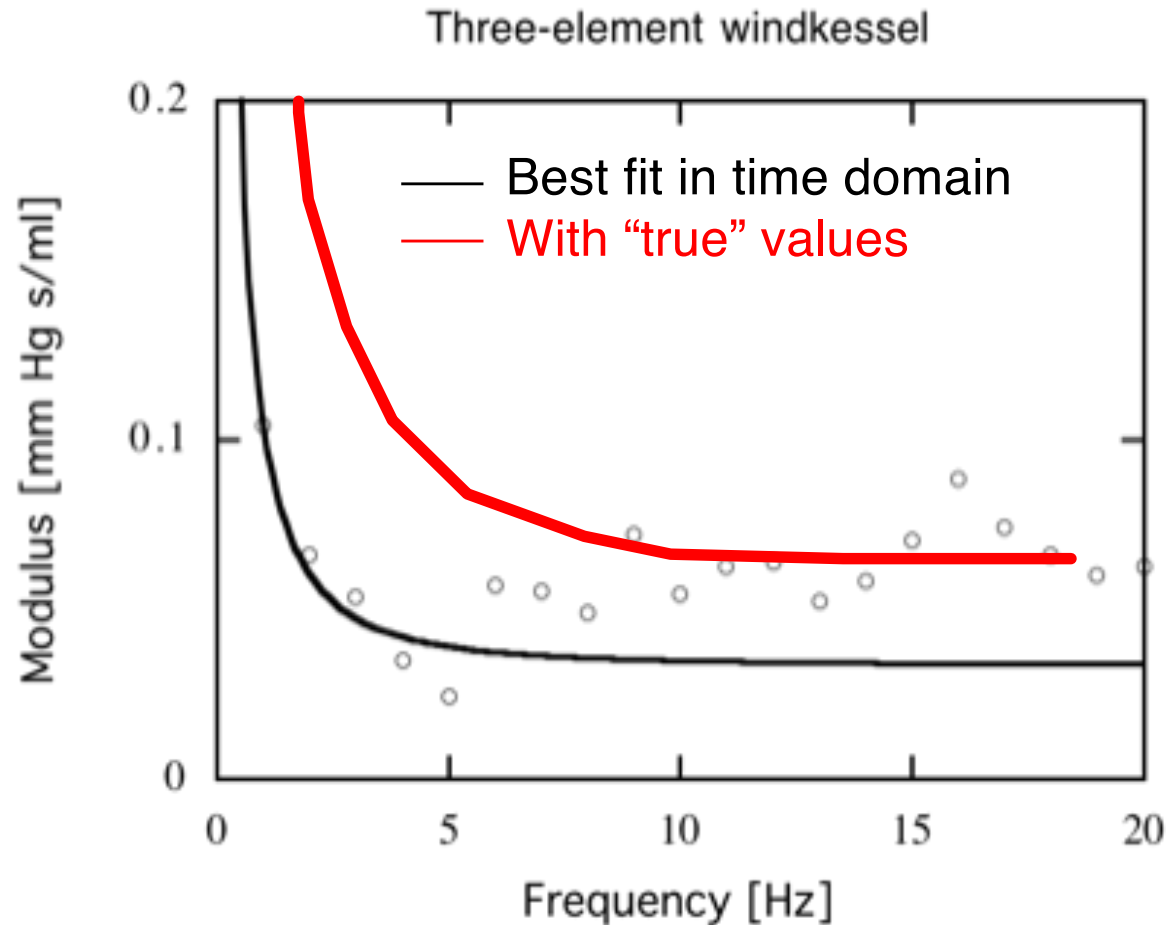
Best fit



With “true” parameters



Why the 3-element model falls short

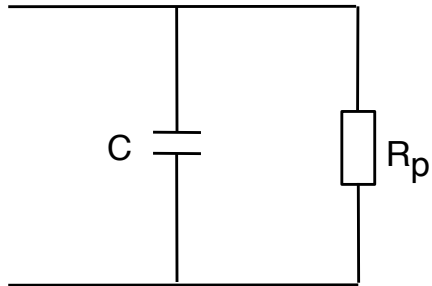


Characteristic impedance forces low-to-mid frequency input impedance to be much higher. Hence in a time domain fit this leads to estimates of:

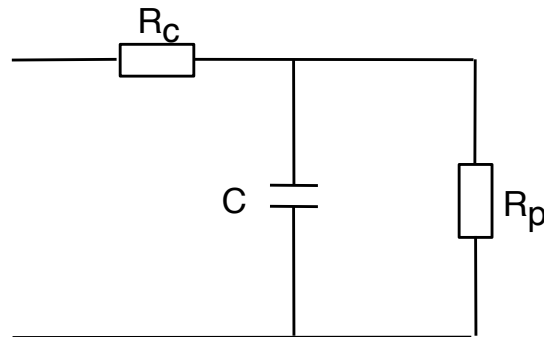
1. Higher compliance
2. Lower characteristic impedance

Windkessel models

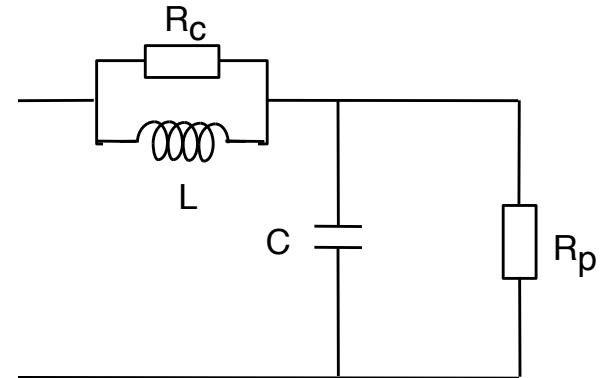
Two-element windkessel



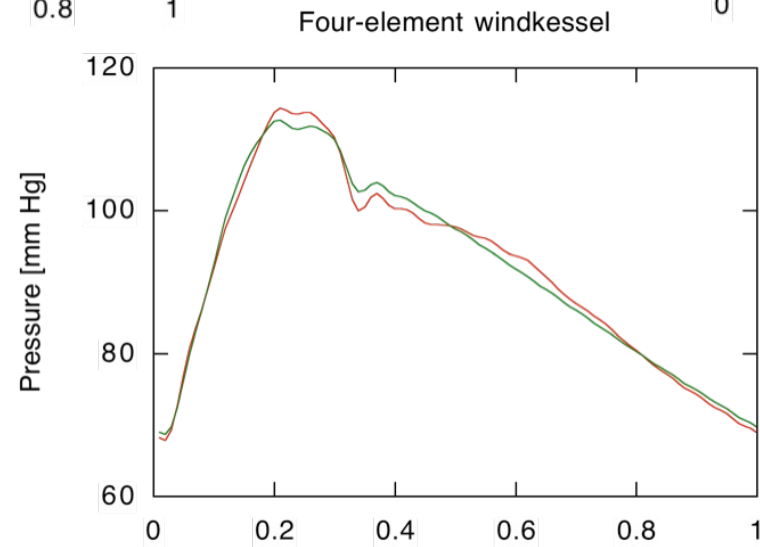
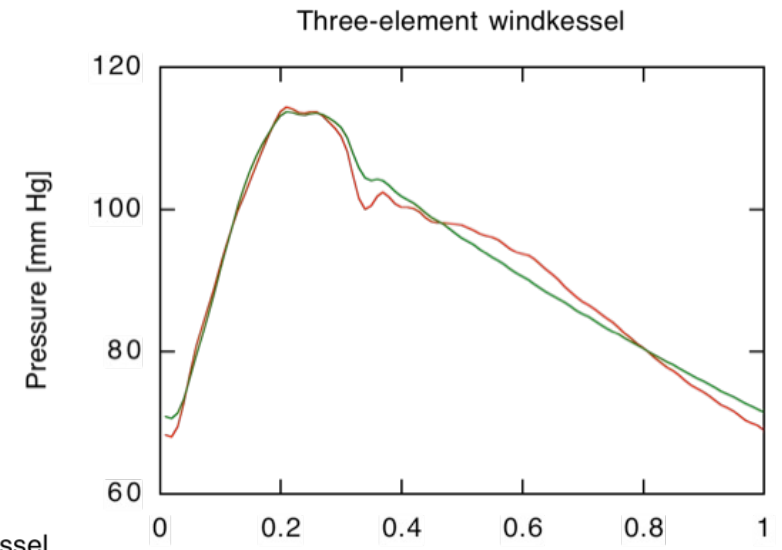
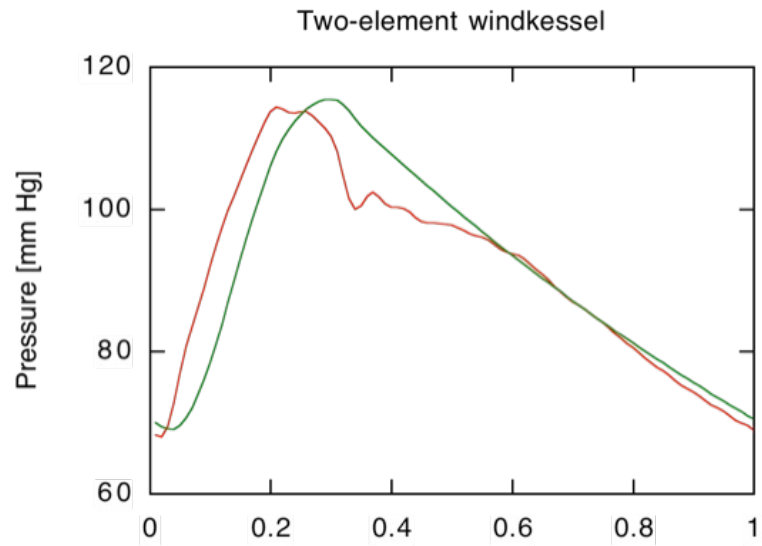
Three-element windkessel



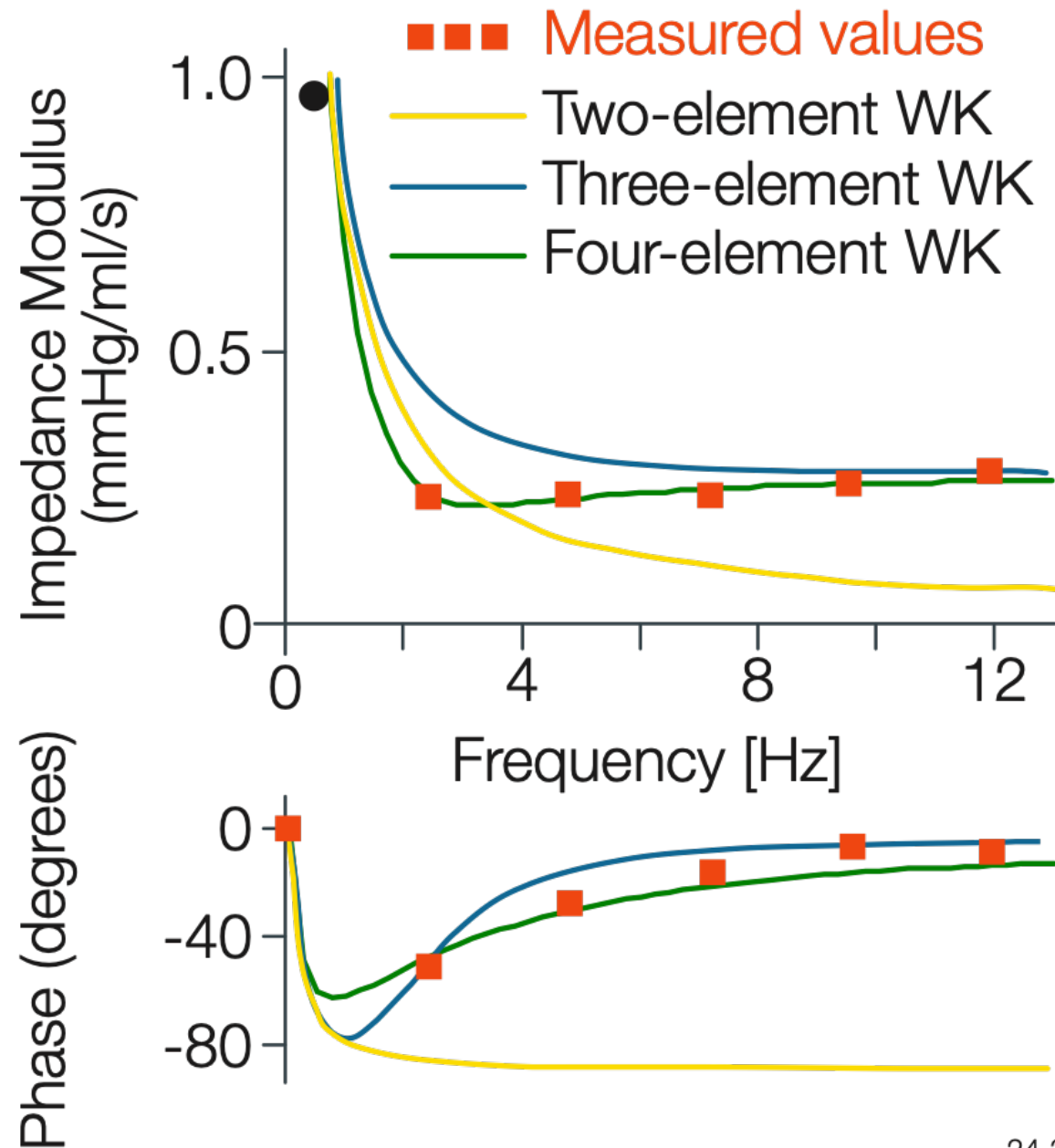
Four-element windkessel



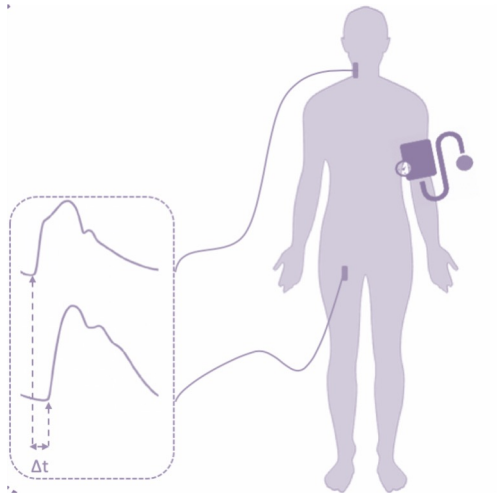
Fit in time domain



Comparison of input impedances



Part II



Use 1-D model to develop inverse methods for the assessment of:

- a) **Cardiac output**
- b) **Ventricular elastance (contractility)**

Using noninvasive peripheral measurements of:

- Brachial pressure
- Pulse wave velocity (carotid to femoral PWV)

A) Cardiac output monitoring

Introduction

Methods

Results

Importance. CO monitoring is essential for patient management.

-> valuable information on global perfusion

Gold standard. Thermo-dilution using a pulmonary artery catheter set the basis for CO monitoring in clinical practice.

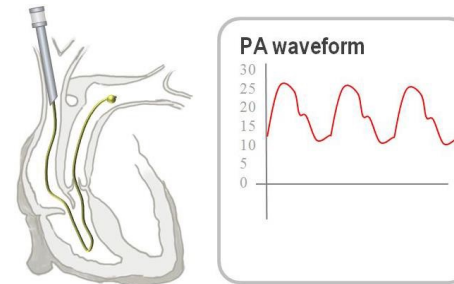


Figure 1. Pulmonary artery catheter positioning and corresponding pressure waveforms in mmHg. Source: www.derangedphysiology.com,
The
Pulmonary Artery Catheter

Noninvasive CO and cSBP prediction

Introduction

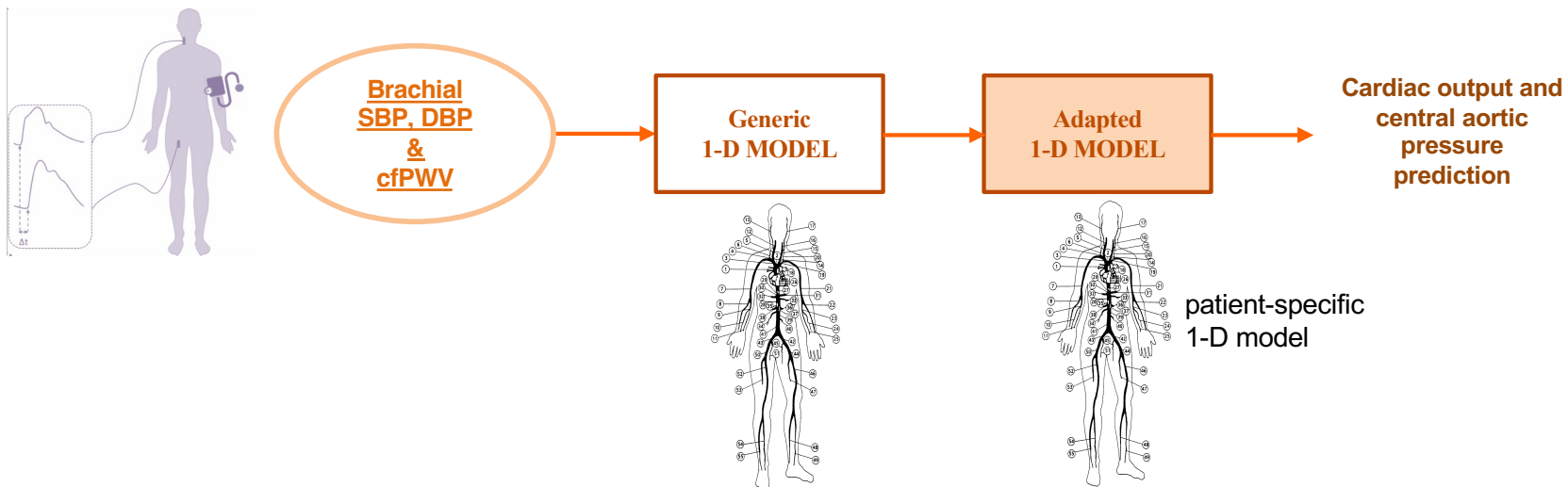
Methods

Results

Inter-subject variability requires theory-based CV models to be as **individualized** as possible (age, gender, hypertension, etc.).

Our approach:

To “tune” a 1-D arterial tree model to patient-specific standards using only noninvasive, easily-obtained peripheral measurement data.



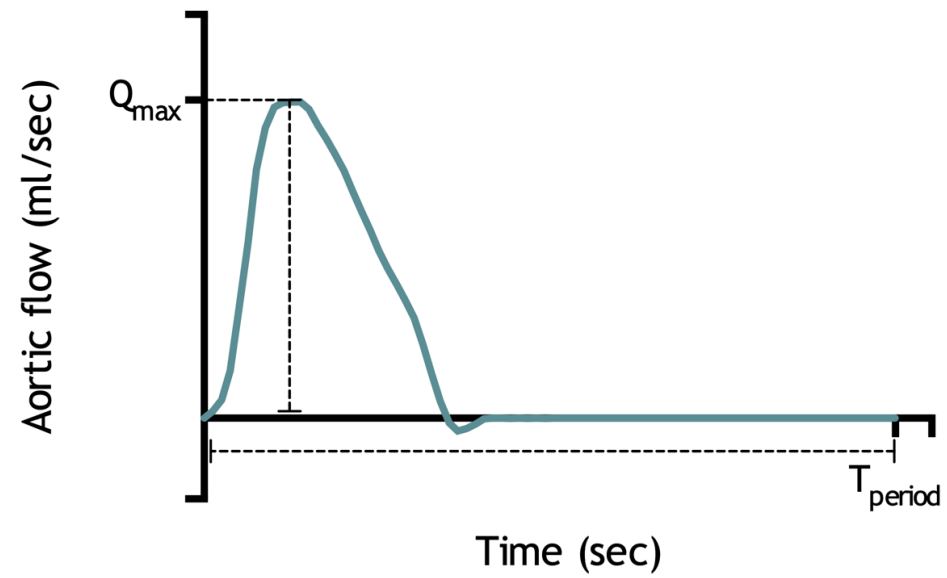
Uncalibrated generic aortic flow

Introduction

Methods

Results

Discussion



Methods

Introduction

Methods

Results

Two-layer optimization algorithm to tune the 1-D model to patient-specific standards

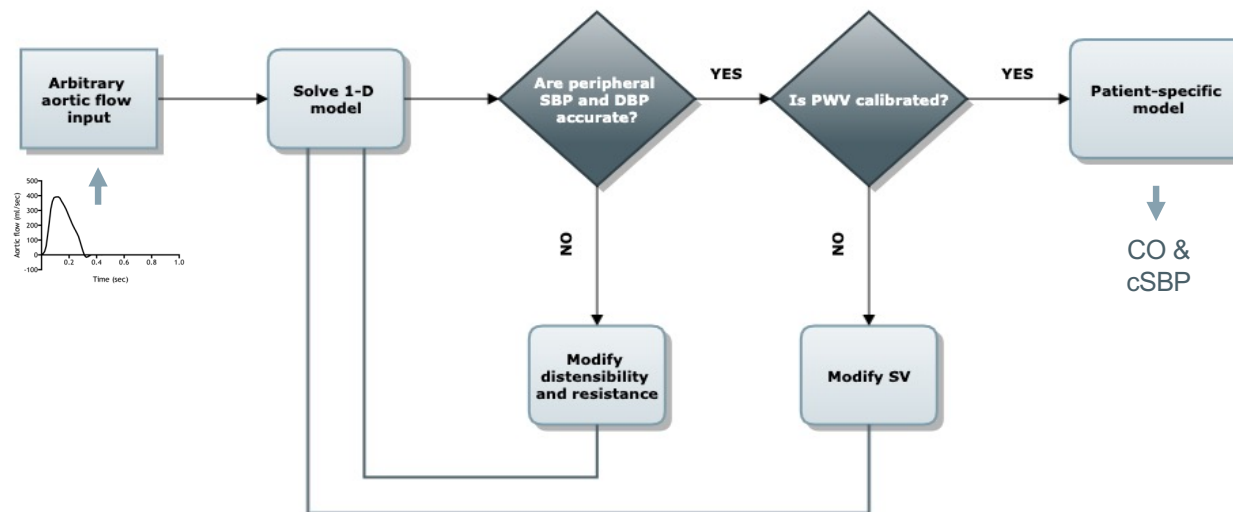


Figure 5. Schematic representation of the two-layer optimization algorithm for predicting noninvasive patient-specific central hemodynamics. SBP: systolic blood pressure, DBP: diastolic blood pressure, PWV: pulse wave velocity, SV: stroke volume.

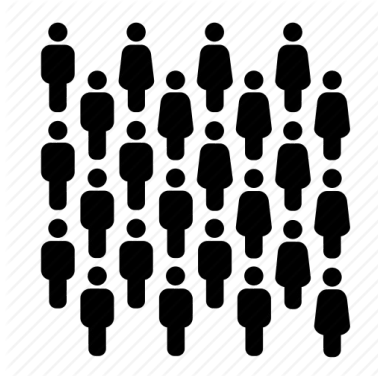
Evaluation

Introduction

Methods

Results

Discussion



20 individuals

TABLE I
DESCRIPTIVE HEMODYNAMIC PARAMETERS AND CLINICAL
CHARACTERISTICS OF THE STUDY POPULATION (N=20)

Index	min	max	mean	SD
Height (cm)	152.00	192.00	173.50	10.39
Weight (kg)	49.00	128.00	78.60	21.63
Central SBP (mmHg)	93.00	136.00	111.00	11.94
Brachial SBP (mmHg)	98.00	145.00	117.55	11.72
Brachial DBP (mmHg)	60.00	98.00	77.25	9.18
Brachial PP (mmHg)	32.00	52.00	40.30	5.17
MAP (mmHg)	72.67	110.33	90.68	9.79
Mean aortic flow (L/min)	3.00	6.20	4.34	0.88
HR (bpm)	51.00	89.00	70.85	10.40
cf-PWV (m/s)	5.25	11.25	6.89	1.92

Measurements

Introduction

Methods

Results

Discussion

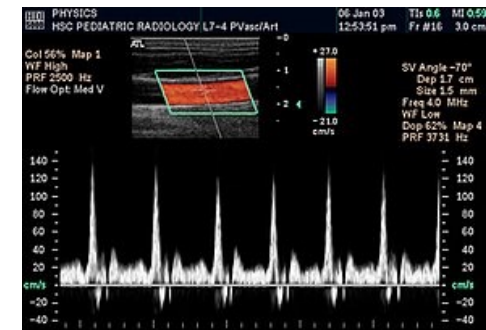
- Mobil-O-Graph peripheral and central pressure data



- cf-PWV from SphygmoC



- Ultrasound 2D central flow



In vivo validation

- **In vivo anonymized data** from n=20 subjects [Papaioannou et al.]

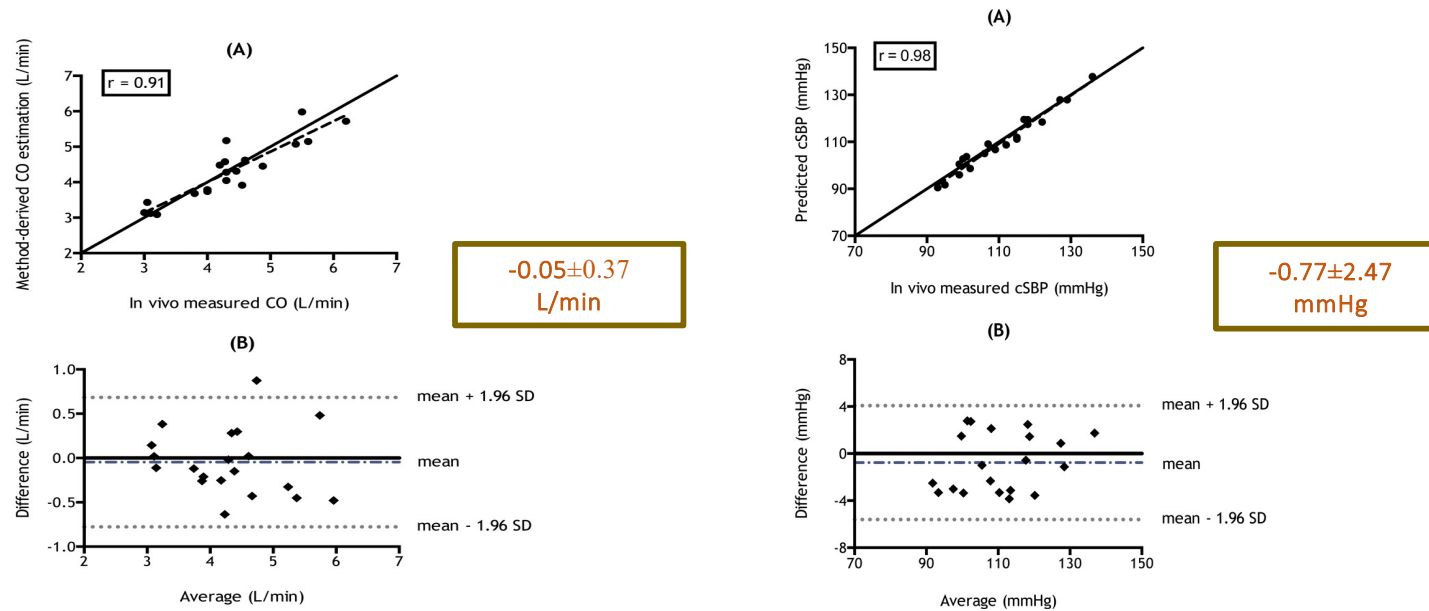


Figure 6. Scatter plot between the actual CO values and the model-derived CO estimates (solid line represents equality).

Figure 7. Scatter plot between the measured cSBP values and the model-derived cSBP estimates (solid line represents equality).

B. Noninvasive assessment of cardiac contractility

Monitoring of cardiac performance is indispensable

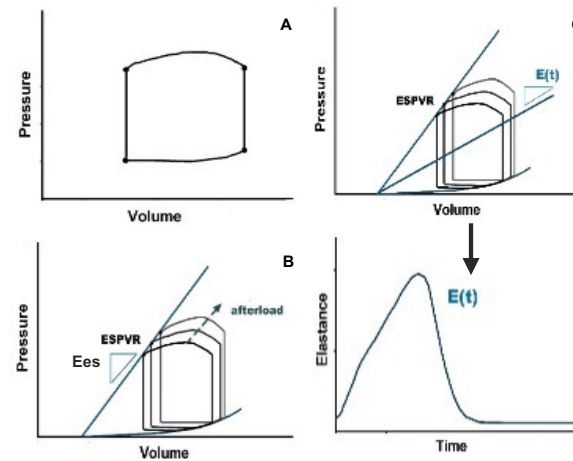
Best index of left ventricular (LV) contractility:

End-systolic
elastance, E_{es}

Measurement of pressure-volume loop is the gold standard for diagnosis of heart disease

However, it is:

- **expensive,**
- **invasive,**
- applicable only to **specific** clinical settings



Noninvasive assessment of cardiac contractility

Our approach

Derive Ees and the entire P-V loop from noninvasive measurements of :

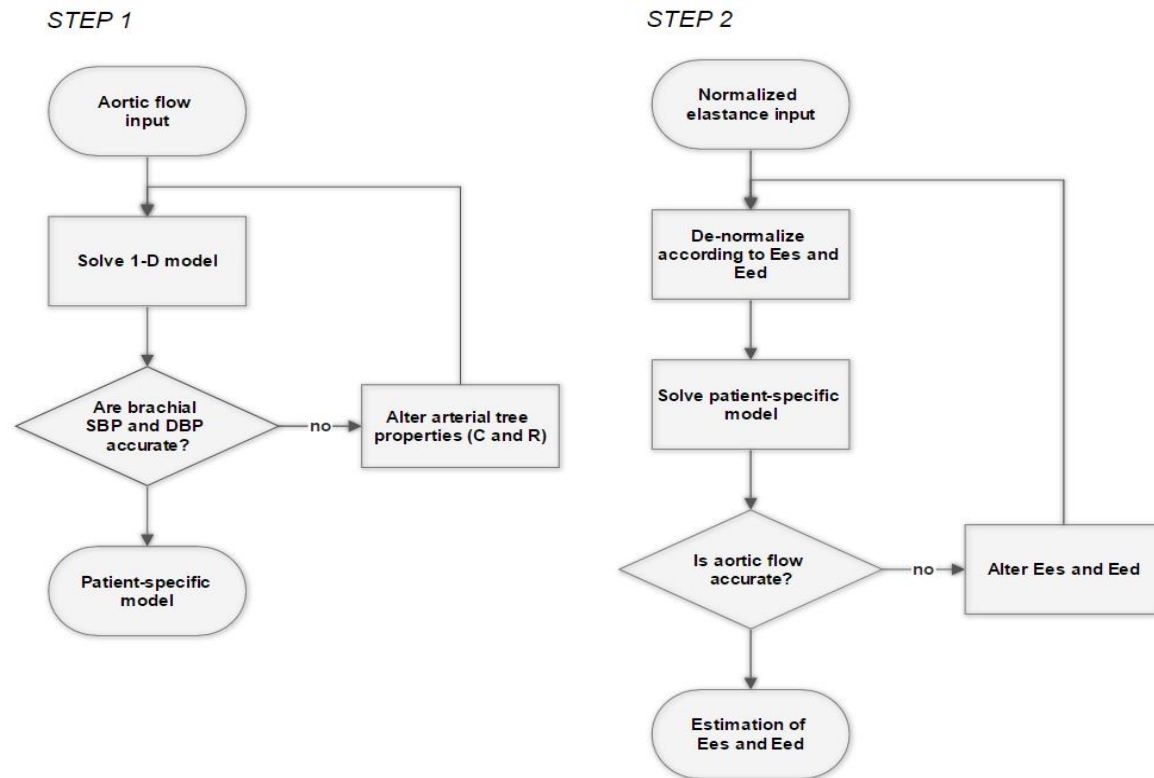
- aortic flow,
- ejection fraction,
- brachial systolic and diastolic pressure

How?

Inverse method in two steps:

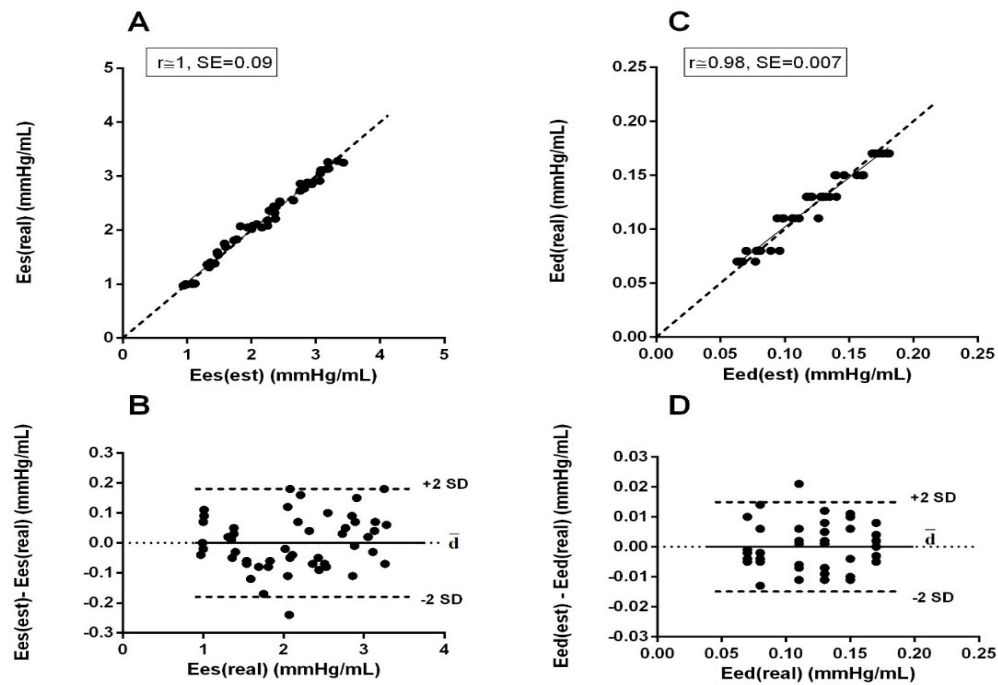
1. Tune of the parameters of the generic 1-D model of the arterial tree (compliance, resistance, geometry, etc) to fit measured pressure
2. Estimation of the Ees and cardiac preload to reproduce measured aortic flow and EF

Method



Results

Validation in a database of virtual patients



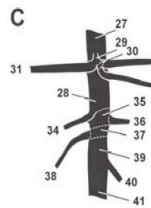
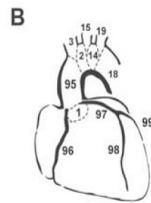
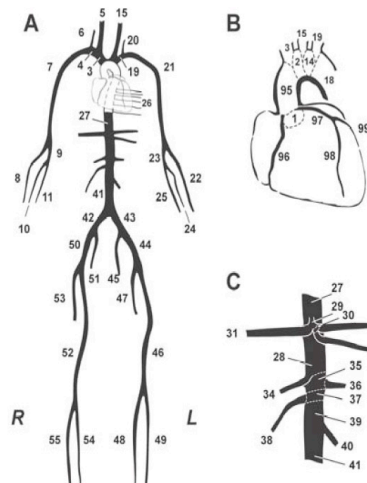
- ✓ Good accuracy
- ✓ Low sensitivity to arterial mistuning and measurement errors

Figure. A & B. Scatterplot and Bland-Altman plot of maximal elastance real vs predicted. C & D. Respective plots for minimal elastance

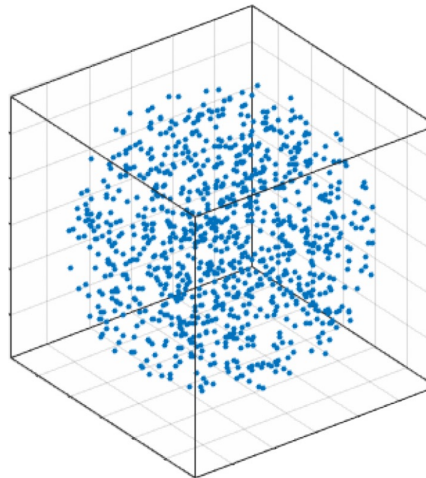


ML approaches for Non-Invasive Cardiovascular Health Monitoring

Generation of Virtual Patients



3818 virtual patients

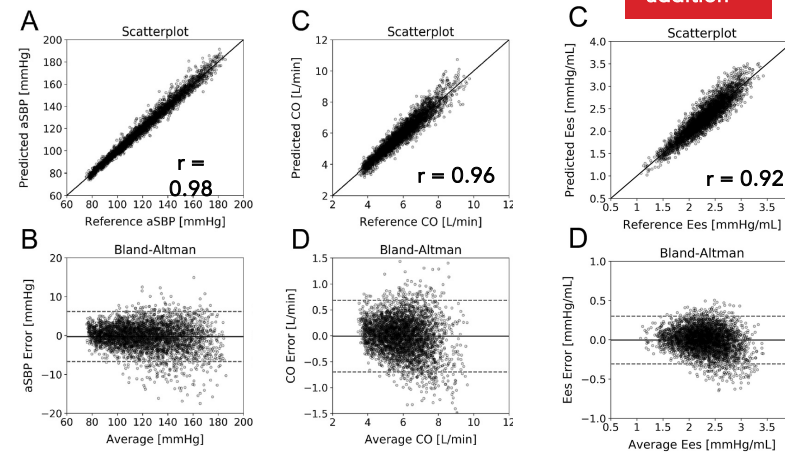
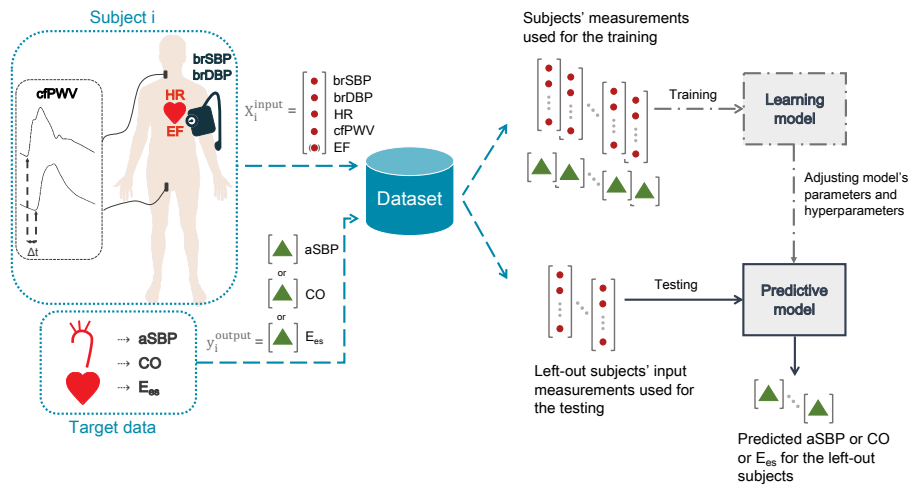


Parameter	<i>In silico</i> population <i>n</i> = 3,818		
	Min	Max	Mean $\pm$ SD
Aortic systolic BP (mmHg)	77	187	123 $\pm$ 24
Aortic diastolic BP (mmHg)	43	128	80 $\pm$ 21
Aortic pulse pressure (mmHg)	11	106	42 $\pm$ 19
Mean arterial pressure (mmHg)	66	155	101 $\pm$ 21
Radial systolic BP (mmHg)	85	193	133 $\pm$ 23
Radial diastolic BP (mmHg)	37	119	73 $\pm$ 21
Radial pulse pressure (mmHg)	21	124	60 $\pm$ 22
Heart cycle (s)	0.6	1	0.7 $\pm$ 0.1
Total peripheral resistance (mmHg·s/mL)	0.6	1.4	1 $\pm$ 0.2
Total arterial compliance (mL/mmHg)	0.3	2.9	1.1 $\pm$ 0.5
Cardiac output (L/min)	3.3	10.5	6 $\pm$ 1.2

Reymond et al., Am. J. Physiol. Heart Circ. Physiol 2009.

ML feature-based CV monitoring

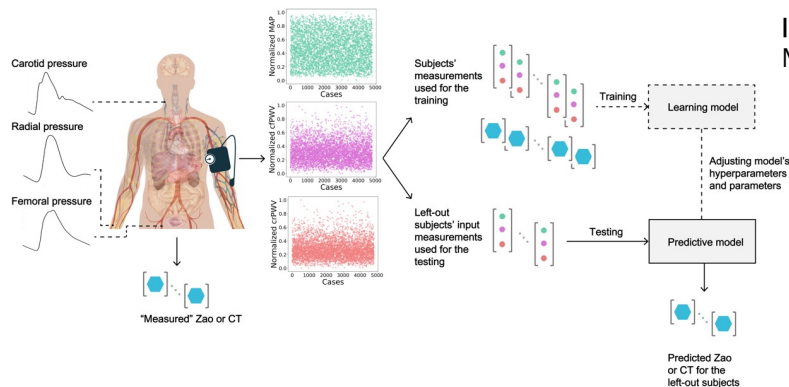
Estimation of aortic hemodynamics and LV end-systolic elastance



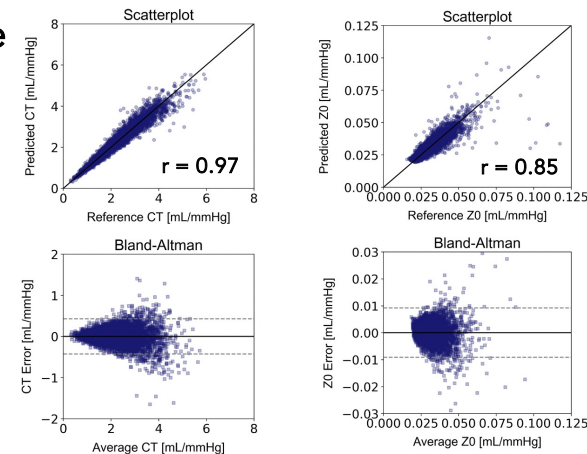
Read the paper



Estimation of total arterial compliance and aortic impedance



Inputs:
MAP, cfPWV, crPWV

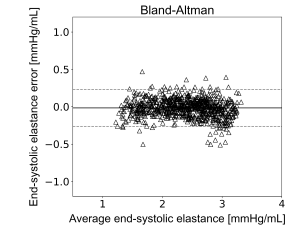
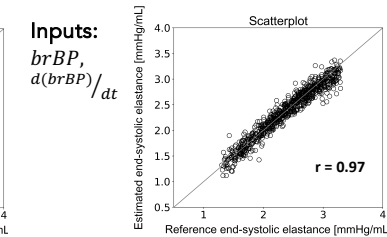
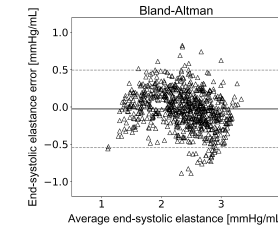
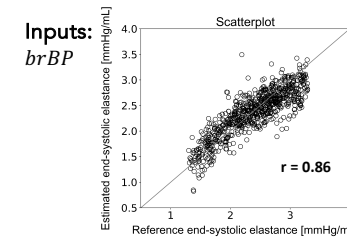
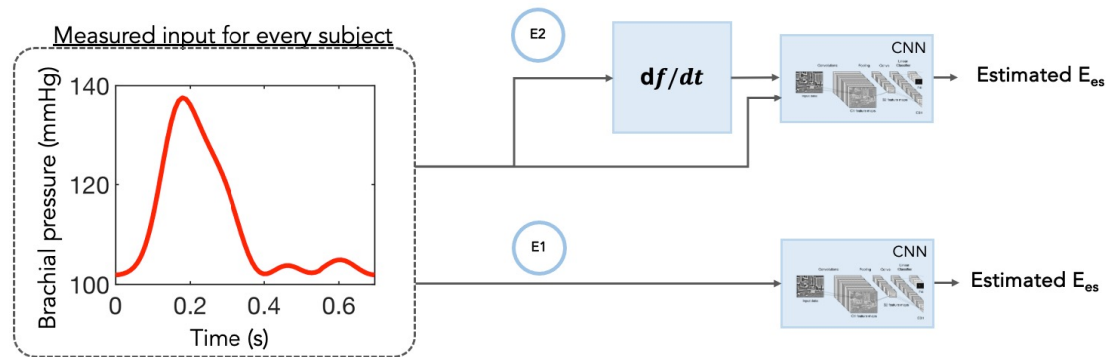


Read the paper

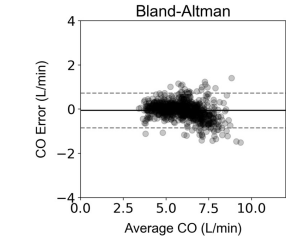
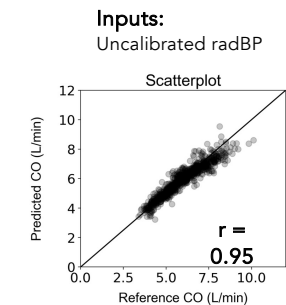
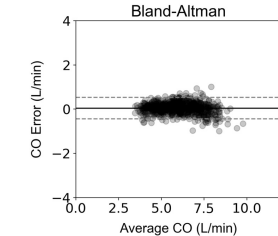
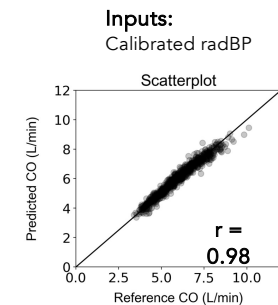
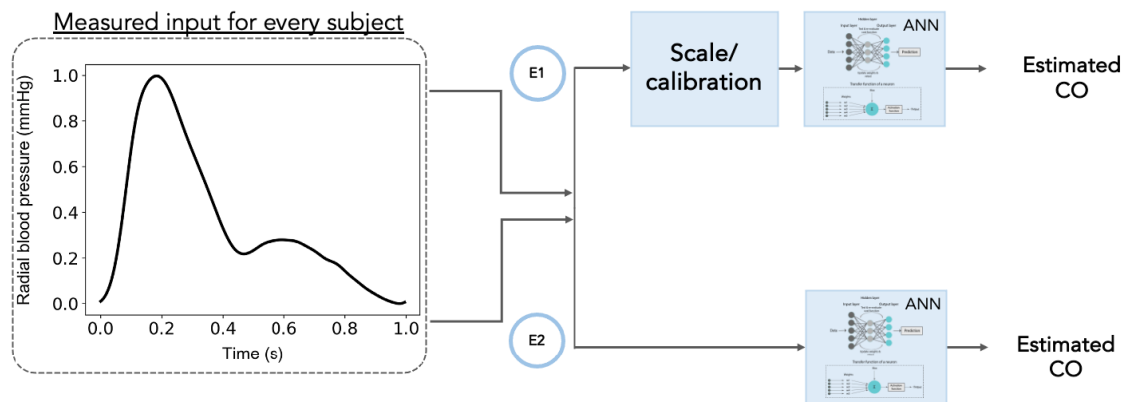


NN-enabled monitoring from raw signals

Estimation of E_{es} using brachial BP waveform via CNN models



Cardiac output estimated from uncalibrated radial BP



*Enjoy VITAL training
...and modeling*

